UNIT 1

Foundation Concepts of AI

CONTENTS

Introduction

What is AI - How AI works- Uses of AI – types of AI - what is Intelligence - Decision Making- How do you make decisions- Introduction to AI and related terminologies: Define AI--Machine Learning -Deep Learning- Differentiate Image processing and computer Vision

Artificial Intelligence

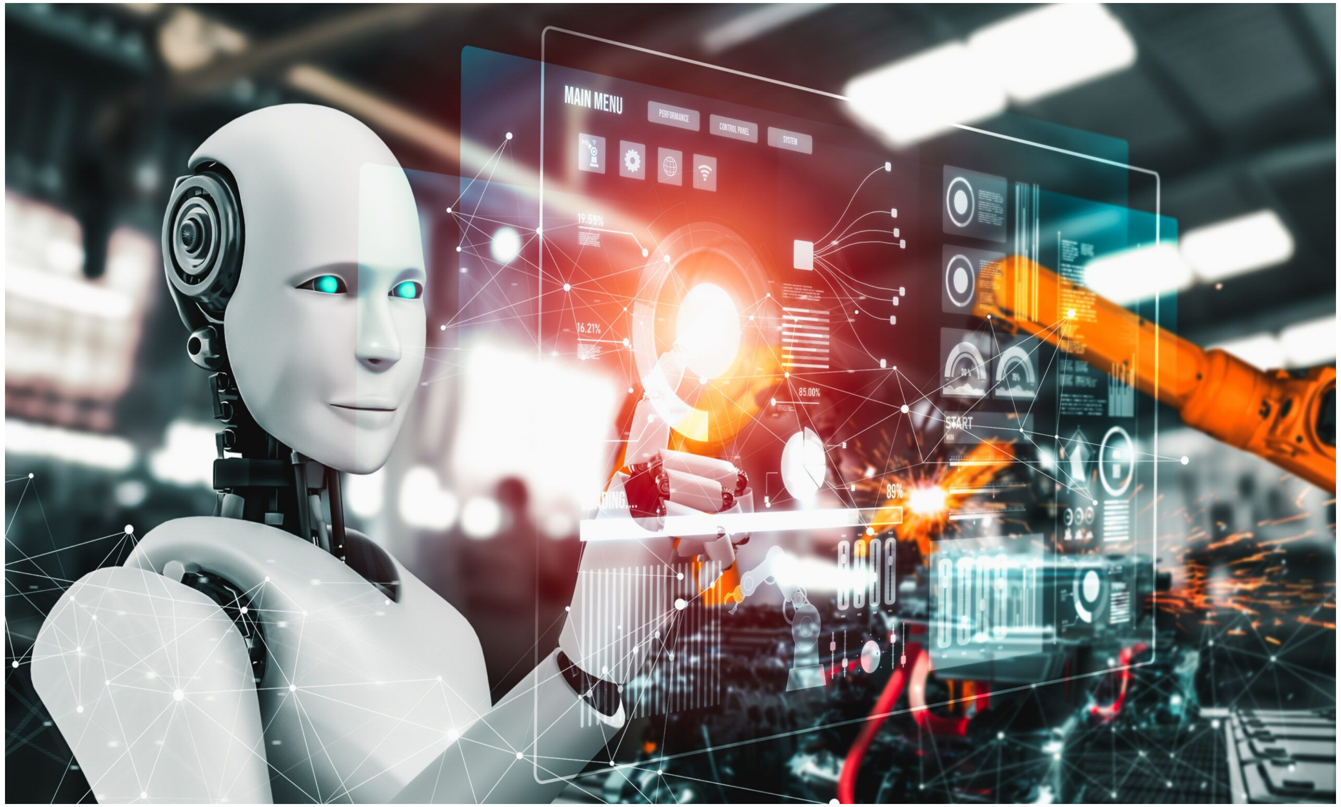
Transforming the Nature of Work, Learning, and Learning to Work

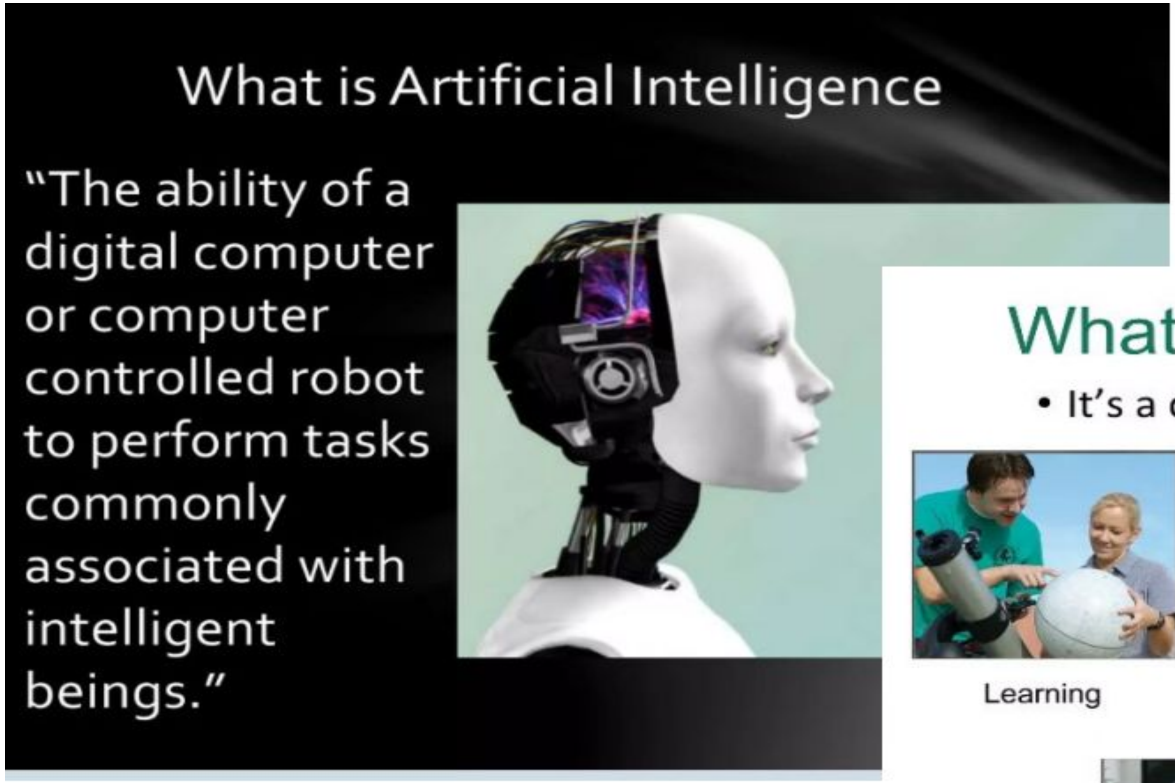
Artificial intelligence (AI) is a popular branch of computer science that concerns with building "intelligent" smart machines capable of performing intelligent tasks.

With rapid advancements in deep learning and machine learning, the tech industry is transforming radically.

What is Artificial Intelligence?

The power of a machine to copy intelligent human behavior
Right???




What is Artificial Intelligence
"The ability of a digital computer or computer controlled robot to perform tasks commonly associated with intelligent beings."


What is Human intelligence?
• It's a composition of abilities like
Learning
Reasoning
Perceiving
Understanding of Language
Feeling

What is Intelligence

- The ability to learn or understand from experience
- The ability to acquire and retain knowledge
- The ability to respond quickly and successfully to a new situation
- The ability to use reason to solve problems

If intelligence is learning, understanding, retaining, responding, and using reason then what is AI?

More Formal Defintion of AI

- AI is a branch of computer science which is concerned with the study and creation of computer systems that exhibit

  Some *form of intelligence*.

  Or

  Those characteristics which we *associate with intelligence in human behavior.*

- It is the *science and engineering* of making intelligent machines, especially intelligent computer programs.

Whats Involved in Intelligence

- Ability to interact with the real world
  - To perceive, understand, and act
    - e.g., speech recognition and understanding
- Searching the best solution
- Reasoning and Planning
  - Modeling the external world – delivery robot
  - Solving new problems, planning, and making decisions
  - Ability to deal with unexpected problems, uncertainties
- Learning and Adaptation
  - We are continuously learning and adapting
  - our internal models are always being “updated”
    - e.g., a baby learning to categorize and recognize animals

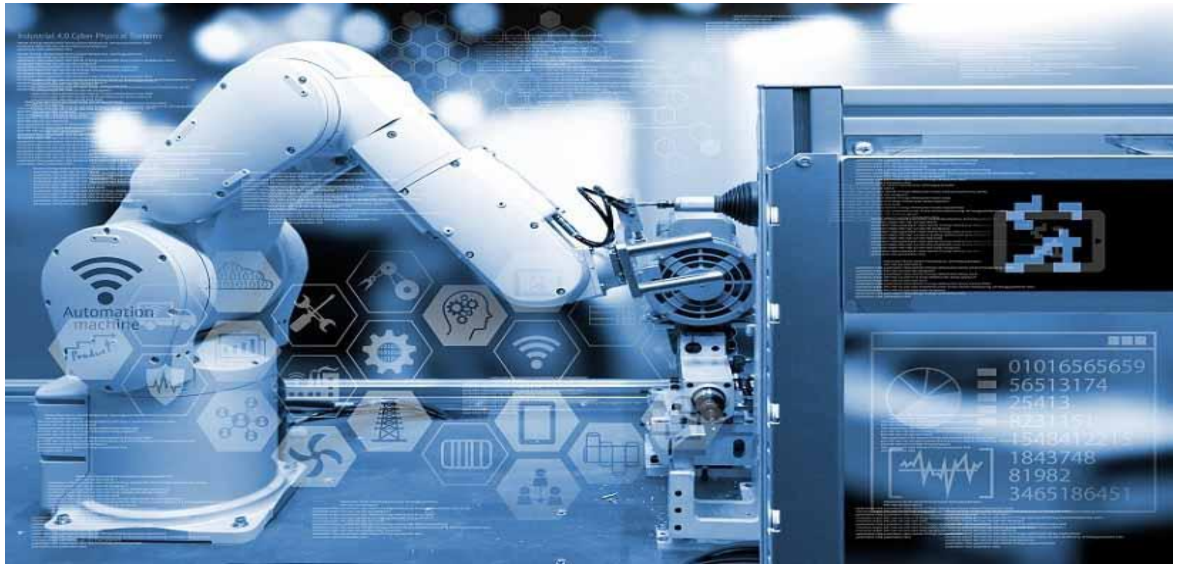
Role of AI

Autonomous Vehicle

Remote patient monitoring

Civil- Construction

EEE- Product defect detection



Smart Factory

Telecommunication - NLP

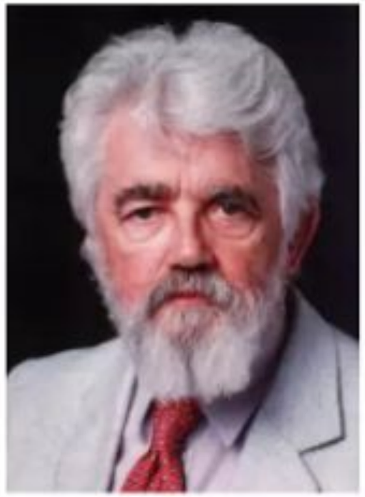
Why Artificial Intelligence ?

- Reduction in Human Error
- zero risks - fully automated production line in a manufacturing facility
- 24x7 Availability - online customer support chatbots
- New Inventions - self-driving cars
- Unbiased Decisions – Resume screening
- Perform Repetitive Jobs – manufacturing (welding, painting, and packaging)
- Daily Applications – Google Map
- AI in Risky Situations
- Pattern Identification
- Medical Applications

Digital Assistance

John McCarthy

- (September 4, 1927 – October 24, 2011) was an American Computer Scientist And Cognitive Scientist.
- McCarthy was one of the founders of the discipline of Artificial Intelligence.
- He coined the term "Artificial Intelligence" (AI)



Goals of AI

Deduction

Drawing specific conclusions from general principles
Premise 1: All humans are mortal.
Premise 2: Socrates is a human.
From these premises, we can deduce the conclusion
Conclusion: Socrates is mortal.

Reasoning

forming judgments based on evidence
Eg:You're planning a trip and need to decide whether to take a train or a bus.
Based on cost, convenience, and travel time you come to conclusion

Problem Solving

process of finding solutions to difficult or complex issues
Eg: chatbot for a customer service application for 24*7

Views of AI fall into four categories in Two dimensions:

Two dimensions

- Thinking/Reasoning vs. Behavior/Action.
- Success according to human standards vs. success according to an ideal concept of intelligence (rationality):

Four Categories.

✓ Systems that think like humans (focus on reasoning and human framework).

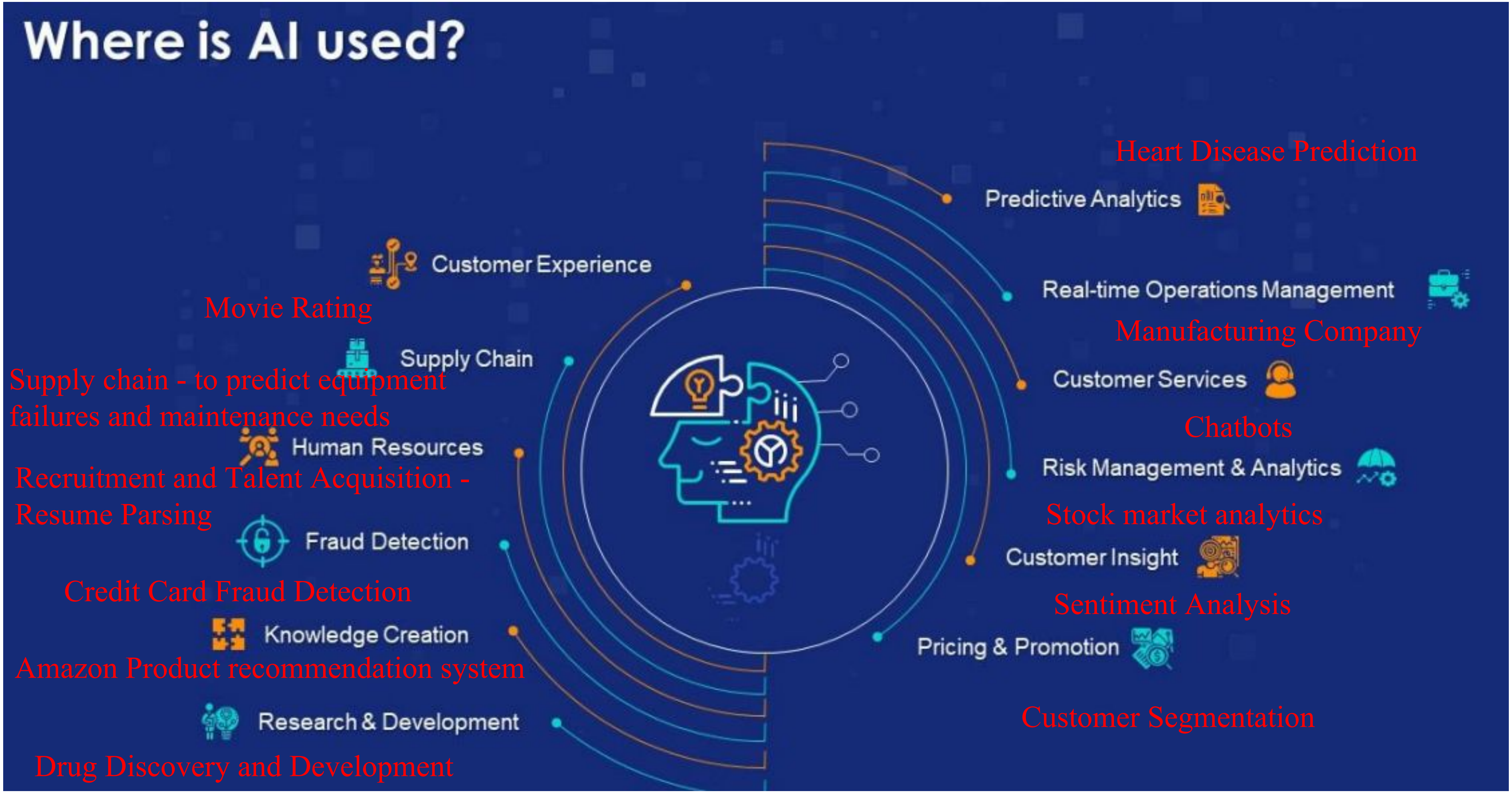
✓ Systems that think rationally (focus on reasoning and a general concept of intelligence).

✓ Systems that act like humans (focus on behavior and human framework).

✓ Systems that act rationally (focus on behavior and a general concept of intelligence).

Artificial Intelligence VS. Robot

AI	Robot
Programmed to think	Programmed to do
Social Interaction	Low level interaction
Learns	Only as smart as program


Where is AI used?
Heart Disease Prediction
Predictive Analytics
Customer Experience
Real-time Operations Management
Movie Rating
Manufacturing Company
Supply Chain
Customer Services
Supply chain - to predict equipment failures and maintenance needs
Chatbots
Human Resources
Risk Management & Analytics
Recruitment and Talent Acquisition - Resume Parsing
Stock market analytics
Fraud Detection
Customer Insight
Credit Card Fraud Detection
Sentiment Analysis
Knowledge Creation
Pricing & Promotion
Amazon Product recommendation system
Research & Development
Customer Segmentation
Drug Discovery and Development

Finally,

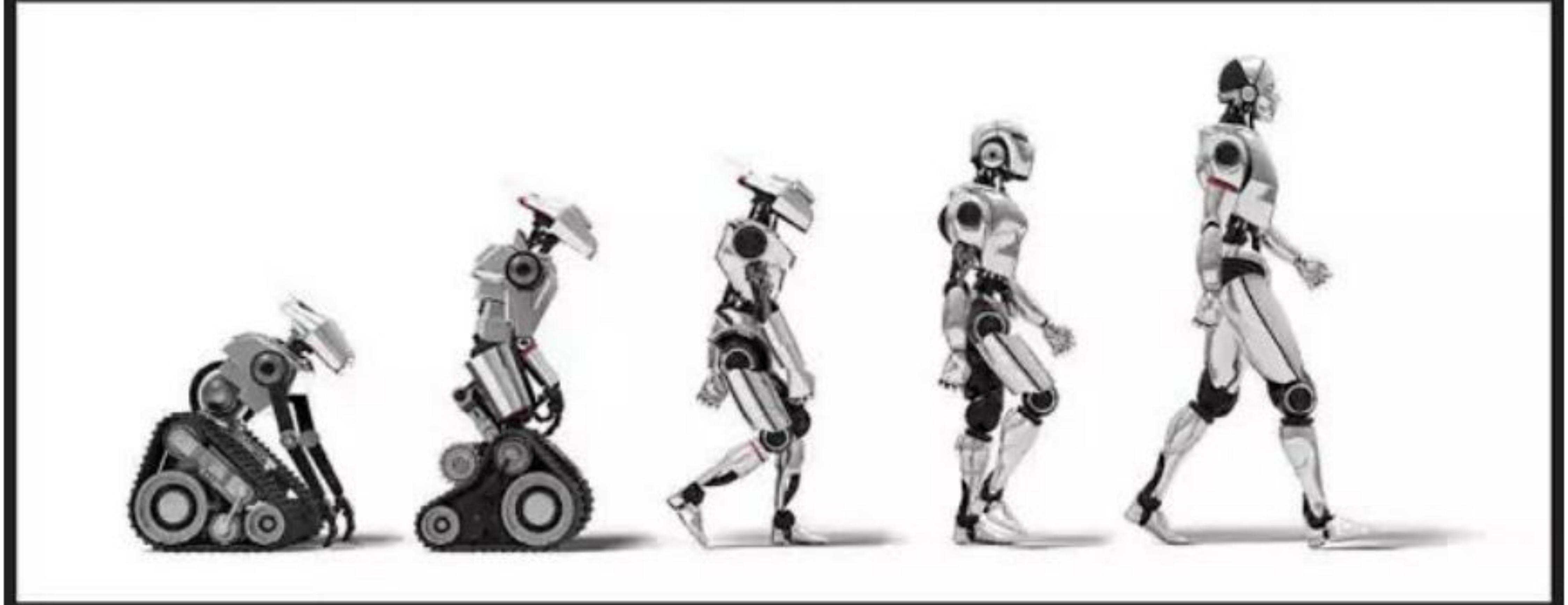
How AI is different????????

Artificial Intelligence	Natural Intelligence
Non Creative	Creative
Precise	May Contain Error
Consistency	Non Consistent
Multitasking	Can't Handle

Where We Are Today?

- Driver-less Transportation
- Automated Assembly Lines and Dangerous Jobs
- Surgery Aid Robots
- Next-Generation Traffic Control

History of Artificial Intelligence -How AI evolved



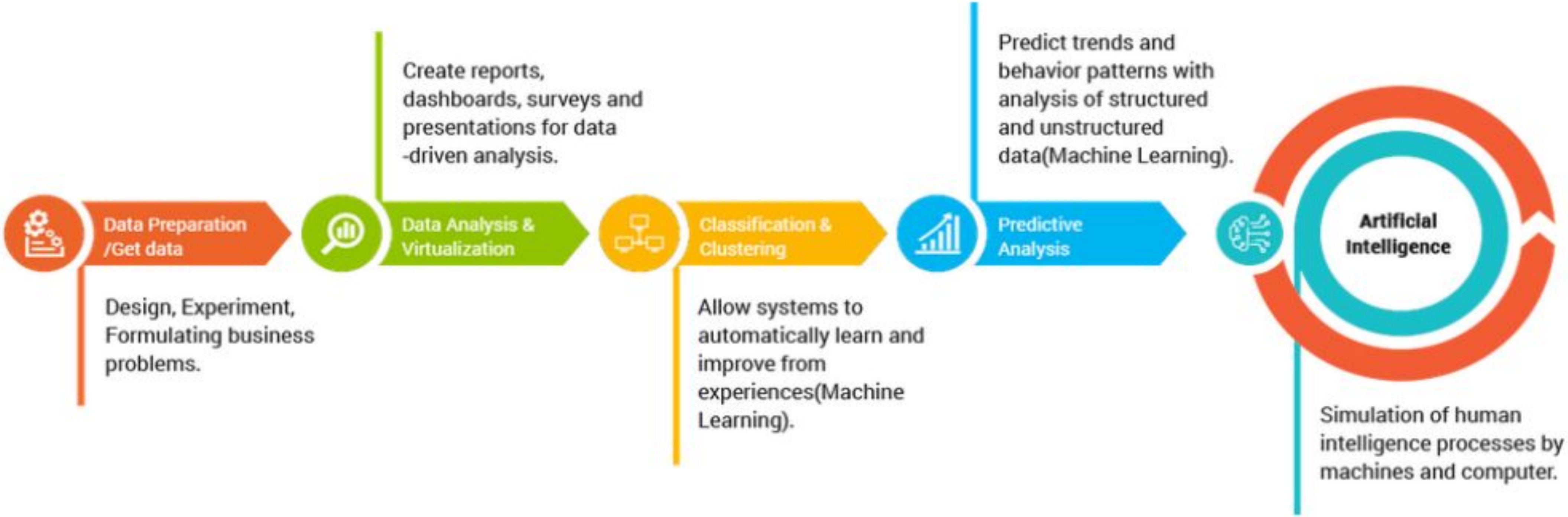
How does AI Works ?

How AI Works

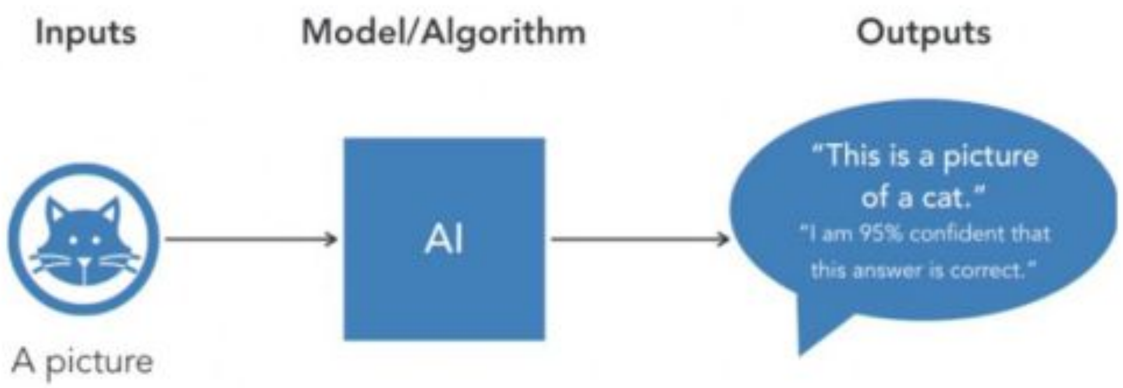
- Formulating Business Problems
- Data Input
  - Various sources such as sensors, databases, text documents, images, videos, or other forms of structured or unstructured data.
- Data Preprocessing
  - Remove noise or inconsistencies - data that do not carry meaningful information
  - Transforming the data into a format suitable for analysis
    - Handling missing value
      - mean, median
  - Reducing the dimensionality of the data.
    - Grouping
      - patterns and relationships between data.

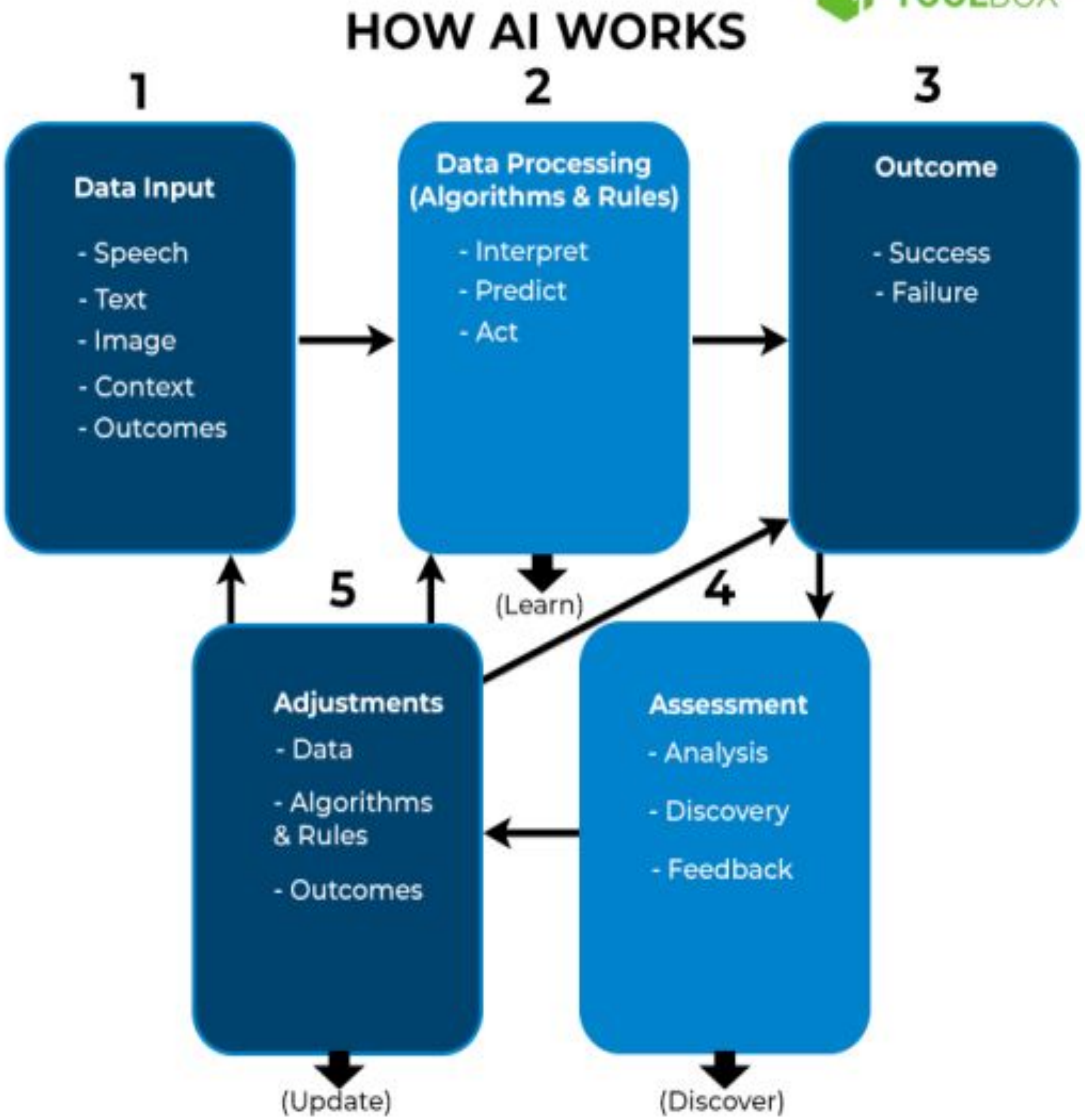
- Data Analysis and Visualization
  - exploring the dataset to understand its structure and contents
  - bar charts or pie charts
  - Uses
    - better understanding of the dataset
    - uncover potential insights
    - identify any data quality
- Building Model
  - Build ML models (Choosing Algorithm based on the problem)
  - Train the model
  - Validation
  - Test the model
- Predictive Analysis
  - to forecast future outcomes

How Artificial Intelligence works ?



AI works by combining large amounts of data with fast, **iterative processing and intelligent algorithms,** allowing the software to learn automatically from patterns or features in the data.


Inputs
Model/Algorithm
Outputs
AI
"This is a picture of a cat."
"I am 95% confident that this answer is correct."
A picture


HOW AI WORKS
1
Data Input
- Speech
- Text
- Image
- Context
- Outcomes
2
Data Processing (Algorithms & Rules)
- Interpret
- Predict
- Act
(Learn)
3
Outcome
- Success
- Failure
4
Assessment
- Analysis
- Discovery
- Feedback
(Discover)
5
Adjustments
- Data
- Algorithms & Rules
- Outcomes
(Update)

What is Machine Learning

- It is a subset of Artificial Intelligence
- Develops algorithms to perform tasks
- Based on learning the hidden patterns of the dataset
- Algorithms are trained on large datasets to learn patterns and make predictions or decisions based on that data

How does Machine Learning Work?

Define Objectives
Identify, the problem to be solved and create a clear objective.

Collect Data
Collect data from hospitals, health insurance companies, social service agencies, police and fire dept.

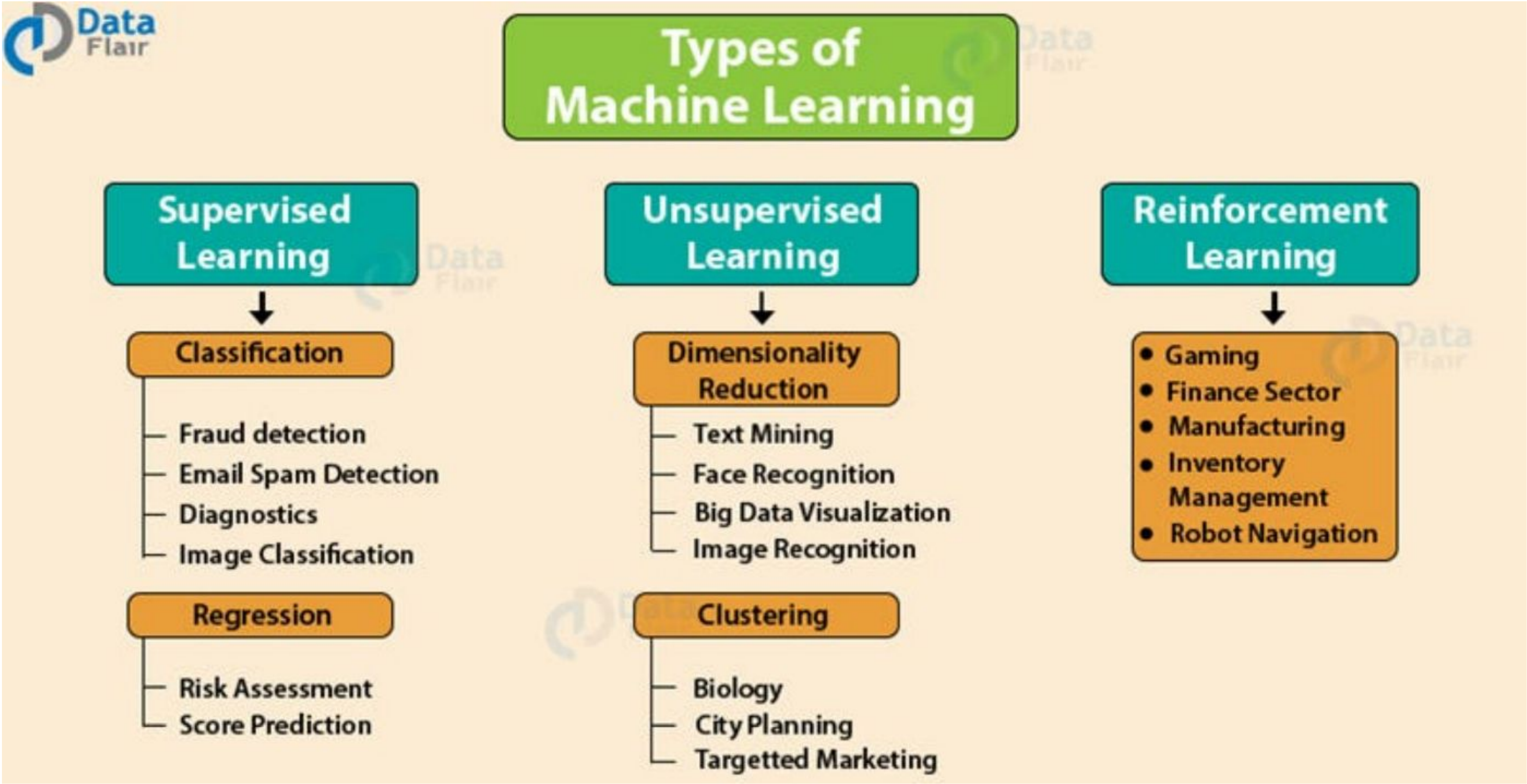
Prepare Data
Preparing data is a crucial step and involves building workflows to clean, match and blend the data.

Select Algorithm
Depend on the problem to be solved and the type of data an appropriate algorithm will be chosen.

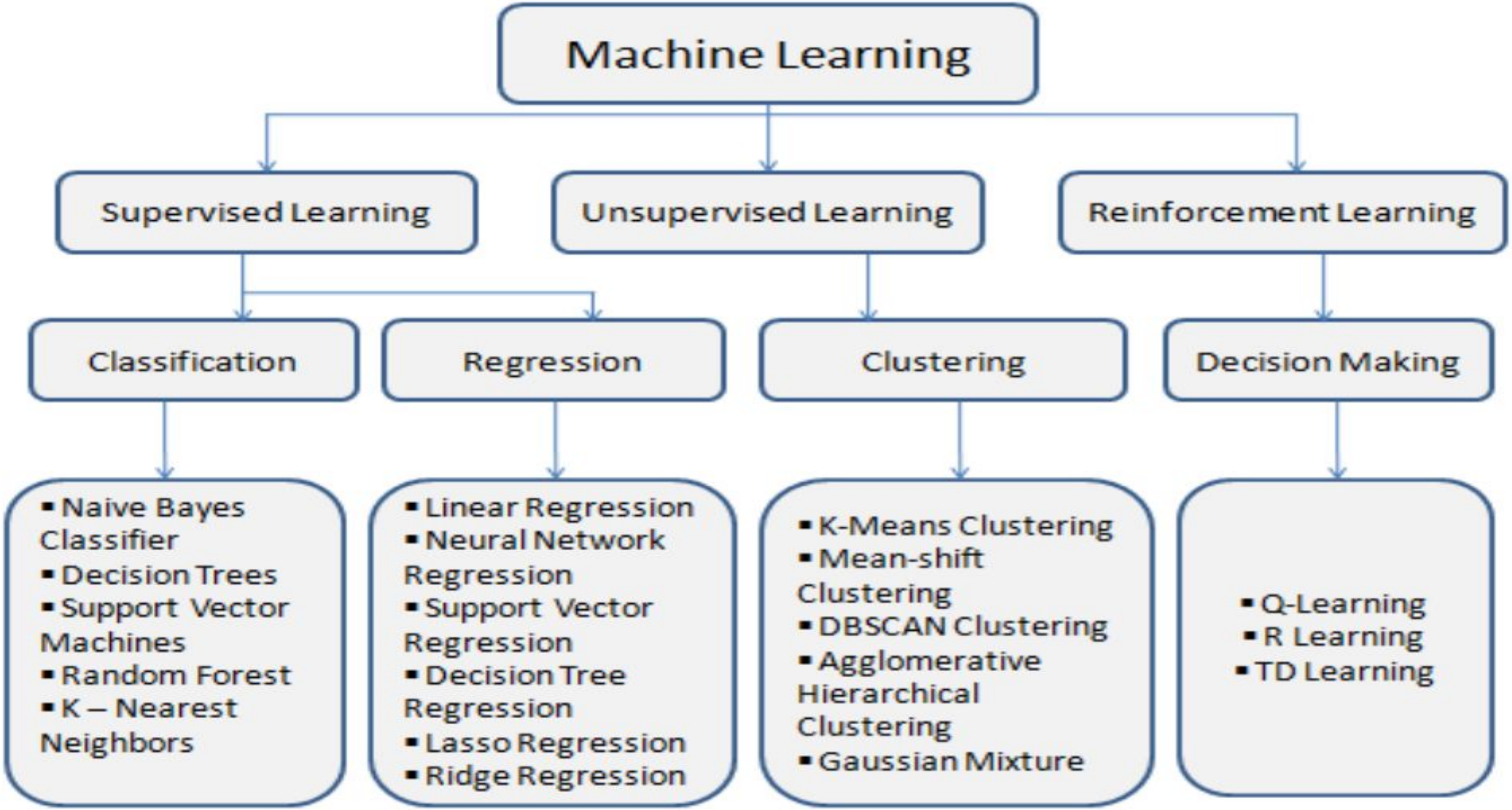
Train Model
Data is fed as input and the algorithm configured with the required parameters. A percent of the data can be utilized to train the model.

Test Model
The remaining data is utilized to test the model, for accuracy. Depending on the results, improvements, can be performed in the "Train model' and/or "Select Algorithm" phases, iteratively.

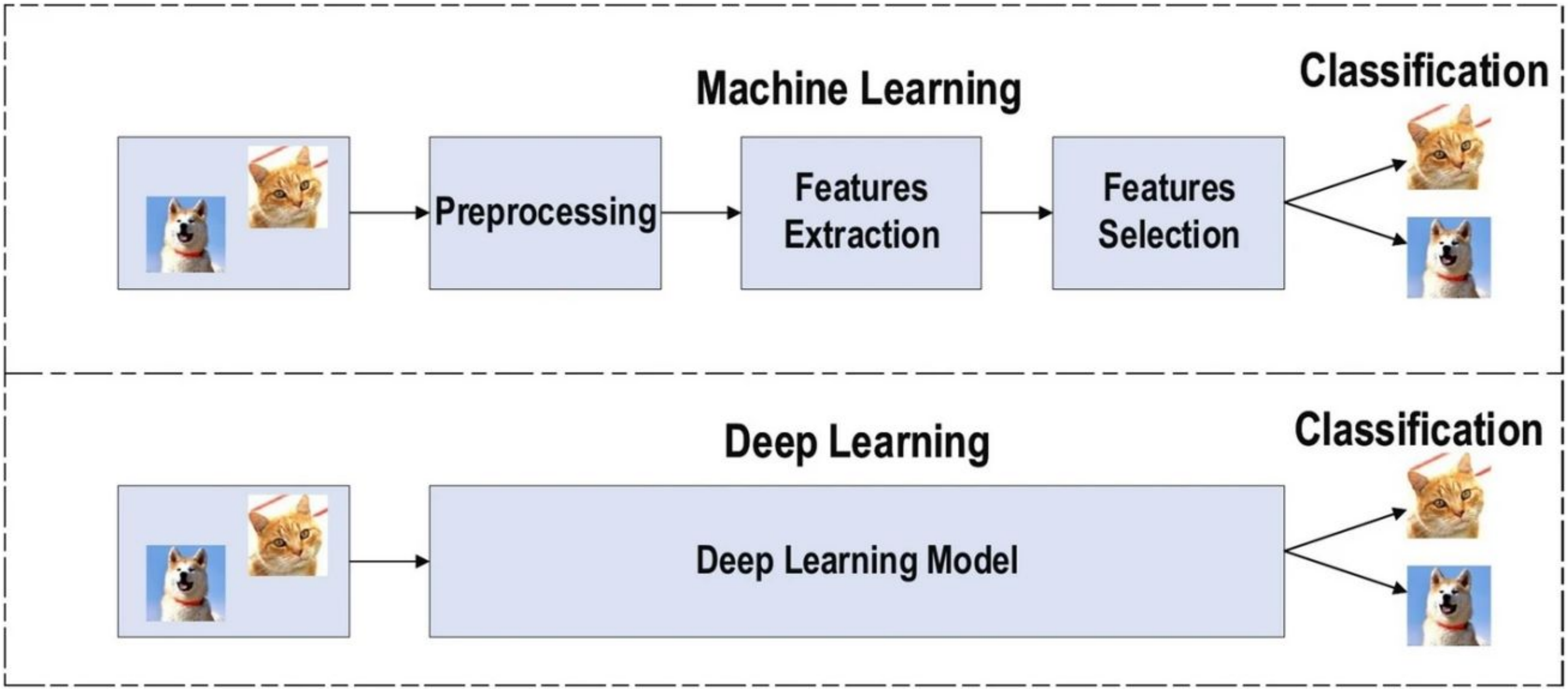
Integrate Model
Publish the prepared experiment as a web service, so applications can use the model.



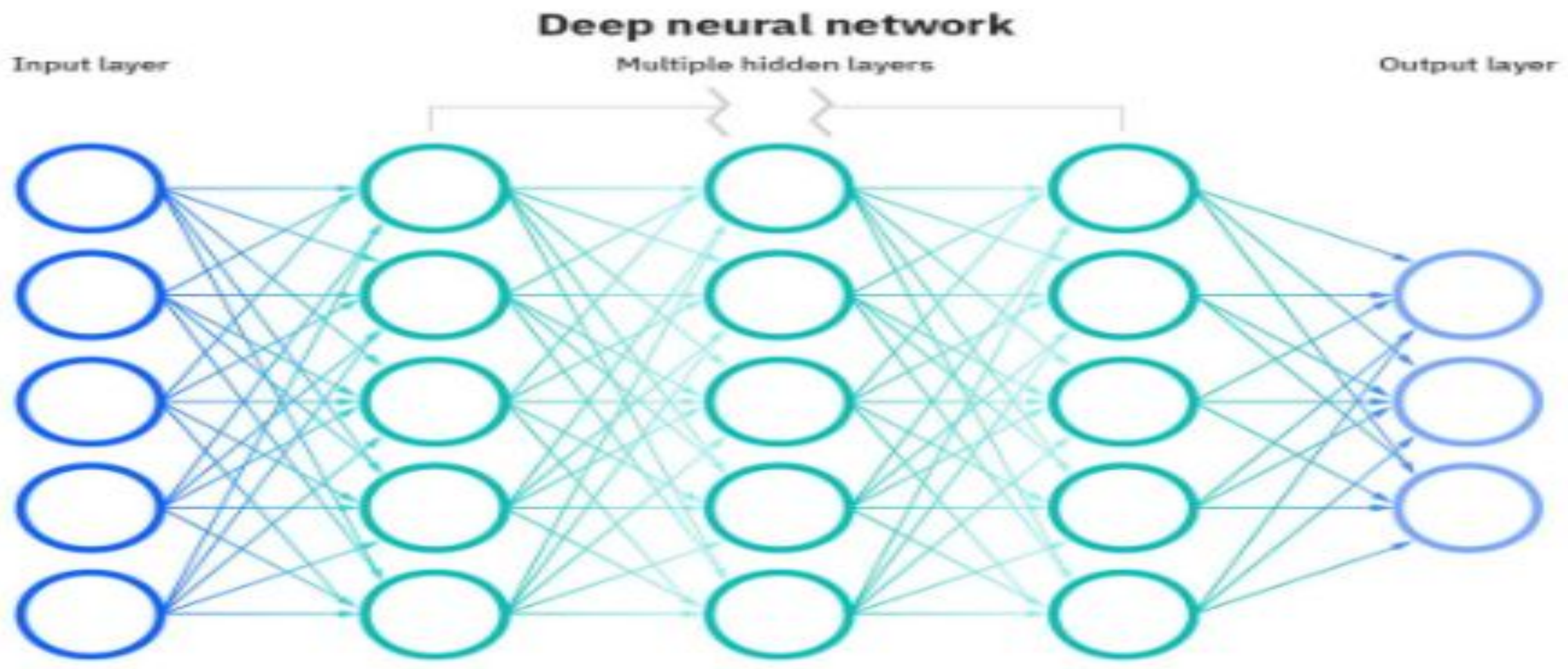
Types of Machine Learning
Supervised Learning
Classification
Fraud detection
Email Spam Detection
Diagnostics
Image Classification
Regression
Risk Assessment
Score Prediction
Unsupervised Learning
Dimensionality Reduction
Text Mining
Face Recognition
Big Data Visualization
Image Recognition
Clustering
Biology
City Planning
Targetted Marketing
Reinforcement Learning
• Gaming
• Finance Sector
• Manufacturing
• Inventory Management
• Robot Navigation

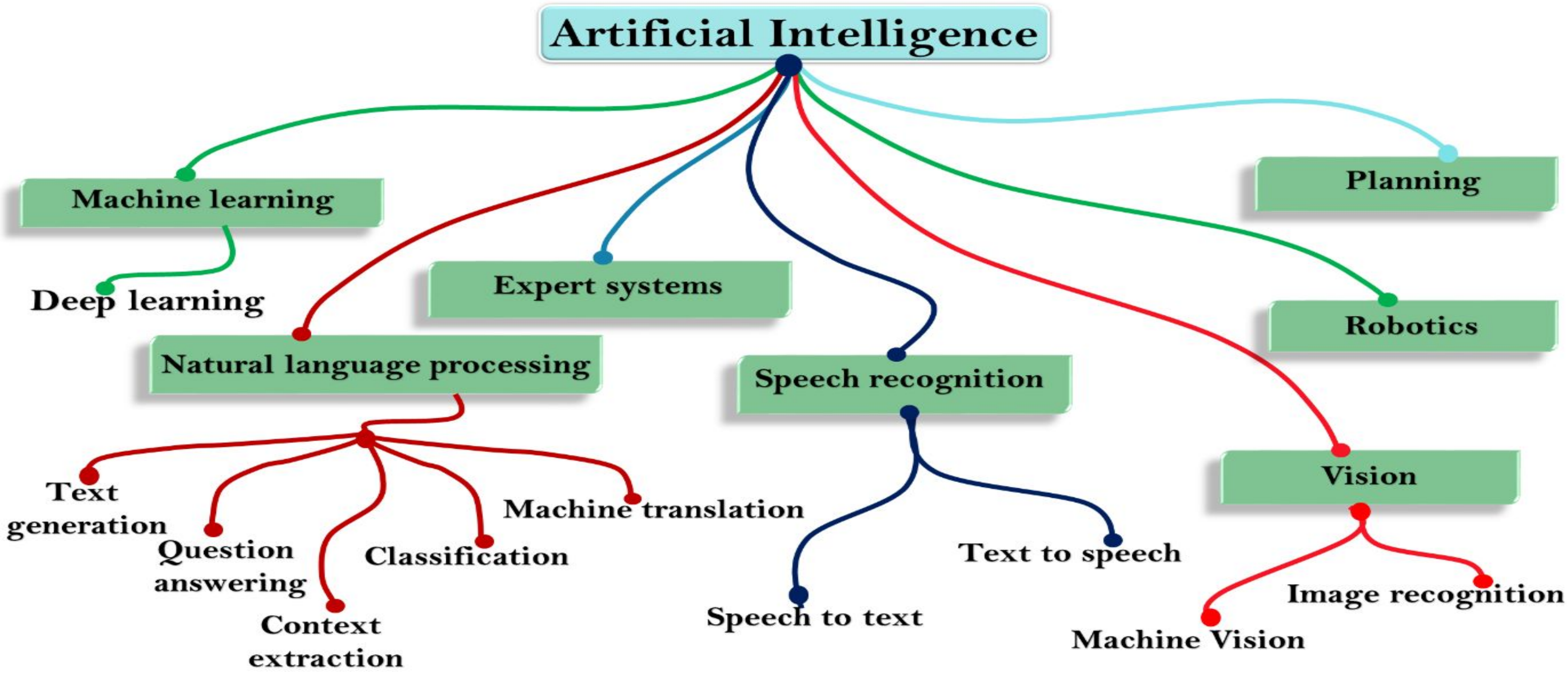

Machine Learning
Supervised Learning
Unsupervised Learning
Reinforcement Learning
Classification
Regression
Clustering
Decision Making
▪ Naive Bayes Classifier
▪ Decision Trees
▪ Support Vector Machines
▪ Random Forest
▪ K – Nearest Neighbors
▪ Linear Regression
▪ Neural Network Regression
▪ Support Vector Regression
▪ Decision Tree Regression
▪ Lasso Regression
▪ Ridge Regression
▪ K-Means Clustering
▪ Mean-shift Clustering
▪ DBSCAN Clustering
▪ Agglomerative Hierarchical Clustering
▪ Gaussian Mixture
▪ Q-Learning
▪ R Learning
▪ TD Learning

Machine Learning vs Deep Learning



Artificial Neural Network





Artificial Intelligence
Machine learning
Deep learning
Expert systems
Natural language processing
Speech recognition
Planning
Robotics
Vision
Text generation
Question answering
Context extraction
Classification
Machine translation
Speech to text
Text to speech
Image recognition
Machine Vision

Unit – 1

Foundation Concepts of AI

What is Intelligence?

Intelligence encompasses multifaceted capacities. It involves learning and comprehension derived from experiences, enabling individuals to acquire and preserve knowledge. Additionally, intelligence manifests in the agility to navigate novel circumstances adeptly, responding swiftly and effectively. Moreover, it entails the application of reasoning to tackle unfamiliar challenges, demonstrating cognitive adaptability and problem-solving prowess. Overall, intelligence encapsulates a dynamic interplay of learning, retention, rapid adaptation, and logical problem-solving, constituting a fundamental aspect of human cognition and behavior.

What is human intelligence?

Human intelligence is the intellectual capability of humans, which is marked by complex cognitive feats and high levels of motivation and self-awareness. Using their intelligence, humans are able to learn, form concepts, understand, and apply logic and reason. Human intelligence is also thought to encompass our capacities to recognize patterns, plan, innovate, solve problems, make decisions, retain information, and use language to communicate.

There are conflicting ideas about how intelligence should be conceptualized and measured. In psychometrics, human intelligence is commonly assessed by intelligence quotient (IQ) tests, although the validity of these tests is disputed. Several subcategories of intelligence, such as emotional intelligence and social intelligence, have been proposed, and there remains significant debate as to whether these represent distinct forms of intelligence. There is also ongoing debate regarding how an individual's level of intelligence is formed, ranging from the idea that intelligence is fixed at birth to the idea that it is malleable and can change depending on a person's mindset and efforts.

What is Artificial Intelligence (AI)?

In Layman terms, Artificial Intelligence can be used defined as a branch of computer science that can simulate human intelligence. AI is implemented in machines to perform tasks that actually require human intelligence. If you are wondering, **what is AI**? Some of their primary function includes the like of reasoning, learning, problem-solving and quick decision making.

At the core of it, Artificial Intelligence is nothing but algorithms with certain sets of rules. AI systems have the ability to learn from the iteration of tasks where the computer data (aka machine learning algorithms) are fed to the system. This is exactly how, machine learning can actually get better at doing their specified tasks, without any external interference.

Artificial Intelligence Vs Machine Learning

Artificial Intelligence is the umbrella term used to mimic human intelligence. On the other hand, Machine Learning (ML) is a subset of Artificial Intelligence (AI) that enables a machine to think independently and make choices without any external influence.

Traditionally, AI was programmed with a certain set of roles based on which they could act. These are formed by simple if-else statements. But, ML enables these devices to act based on the data they collect.

Why Artificial Intelligence?

- Reduction in Human Error
- zero risks - fully automated production line in a manufacturing facility
- 24x7 Availability - online customer support chatbot’s
- New Inventions - self-driving cars
- Unbiased Decisions – Resume screening
- Perform Repetitive Jobs – manufacturing (welding, painting, and
- packaging)
- Daily Applications – Google Map
- AI in Risky Situations
- Pattern Identification
- Medical Applications

Goals of AI

Deduction:

Deduction in AI refers to the process of deriving logical conclusions from given premises or information. Through deductive reasoning, AI systems can infer new truths based on established facts or rules. For instance, in a medical diagnosis system, if the symptoms match a specific disease profile and the patient fits the demographic criteria, the AI deduces the likely diagnosis, aiding in medical decision-making.

Reasoning:

Reasoning in AI involves the ability to make sense of information, draw inferences, and evaluate possibilities to reach logical conclusions. AI employs various forms of reasoning, including inductive, abductive, and analogical reasoning. For example, in autonomous vehicles, reasoning algorithms analyze sensory data, traffic rules, and contextual cues to make informed decisions, such as when to accelerate, brake, or change lanes.

Problem Solving:

Problem solving in AI refers to the capability to devise effective strategies to overcome obstacles or achieve desired outcomes in diverse contexts. AI systems utilize algorithms and heuristics to explore solution spaces, evaluate alternatives, and optimize decision-making. For instance, in logistics optimization, AI algorithms efficiently allocate resources, plan routes, and schedule deliveries to minimize costs and maximize efficiency, addressing complex logistical challenges in real-time.

How AI Works

- **Formulating Business Problems:**
  Formulating business problems in AI involves identifying challenges or opportunities within an organization that can be addressed or leveraged using artificial intelligence techniques. This process entails defining clear objectives, understanding data requirements, and framing problems in a way that enables AI systems to generate effective solutions or insights.
- **Data Input**
  1. Various sources such as sensors, databases, text documents, images, videos, or other
  2. forms of structured or unstructured data.
- **Data Preprocessing**
  1. Remove noise or inconsistencies - data that do not carry meaningful information
  2. Transforming the data into a format suitable for analysis
     - Handling missing value
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  3. Reducing the dimensionality of the data.
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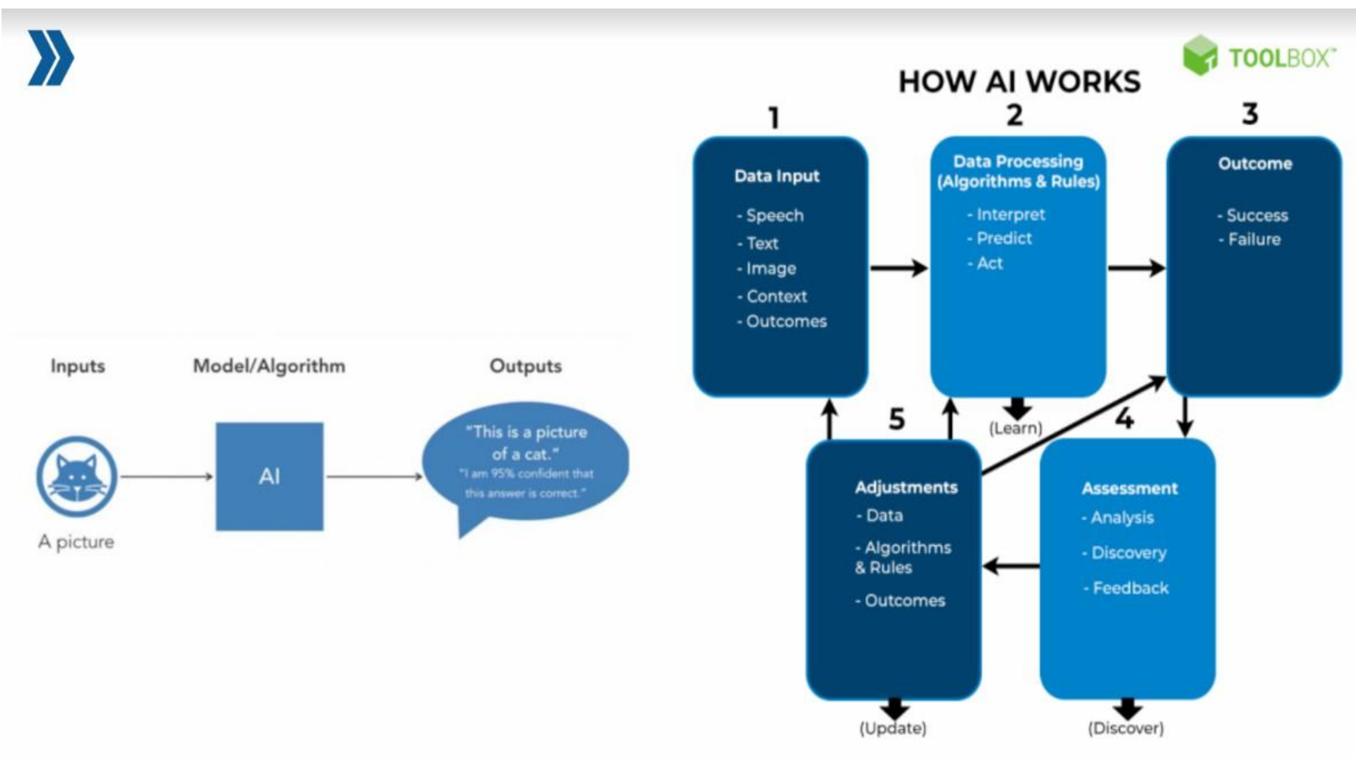


Types of AI

Type 1: Based on Capabilities of AI

1.Narrow AI

Narrow AI is designed and trained on a specific task or a narrow range tasks. These Narrow AI systems are designed and trained for a purpose. These Narrow systems performs their designated tasks but mainly lack in the ability to generalize tasks.

Examples:

Voice assistants like Siri or Alexa that understand specific commands.

Facial recognition software used in security systems.

Recommendation engines used by platforms like Netflix or Amazon.

Despite being highly efficient at specific tasks, Narrow AI lacks the ability to function beyond its predefined scope. These systems do not possess understanding or awareness

2.General AI

General AI refers to AI systems that have human intelligence and abilities to perform various tasks. Systems have capability to understand, learn and apply across a wide range of tasks that are similar to how a human can adapt to various tasks.

While General AI remains a theoretical concept, researchers aim to develop AI systems that can perform any intellectual task a human can. It requires the machine to have consciousness, self-awareness, and the ability to make independent decisions, which is not yet achievable.

Potential Applications:

Robots that can learn new skills and adapt to unforeseen challenges in real-time.

AI systems that could autonomously diagnose and solve complex medical issues across various specializations.

3.Super AI

Super AI surpasses intelligence of human in solving-problem, creativity, and overall abilities.

Super AI develops emotions, desires, need and beliefs of their own. They are able to make decisions of their own and solve problem of its own. Such AI would not only be able to complete tasks better than humans but also understand and interpret emotions and respond in a human-like manner.

While Super AI remains speculative, it could revolutionize industries, scientific research, and problem-solving, possibly leading to unprecedented advancements. However, it also raises ethical concerns regarding control and regulation.

Type 2: Based on the Functionality of AI

Reactive Machines : Reactive machines are the most basic form of AI. They operate purely based on the present data and do not store any previous experiences or learn from past actions. These systems respond to specific inputs with fixed outputs and are unable to adapt.

Examples:

IBM's Deep Blue, which defeated the world chess champion Garry Kasparov in 1997. It could identify the pieces on the board and make predictions but could not store any memories or learn from past games.

Google's AlphaGo, which played the board game Go using a similar approach of pattern recognition without learning from previous games.

Limited Memory AI : Limited Memory AI can learn from past data to improve future responses. Most modern AI applications fall under this category. These systems use historical data to make decisions and predictions but do not have long-term memory. Machine learning models, particularly in autonomous systems

and robotics, often rely on limited memory to perform better.
Examples:
Self-driving cars: They observe the road, traffic signs, and movement of nearby cars, and make decisions based on past experiences and current conditions.
Chatbots that can remember recent conversations to improve the flow and relevance of replies.

Theory of Mind : Theory of Mind AI aims to understand human emotions, beliefs, intentions, and desires. While this type of AI remains in development, it would allow machines to engage in more sophisticated interactions by perceiving emotions and adjusting behavior accordingly.
Potential Applications:
Human-robot interaction where AI could detect emotions and adjust its responses to empathize with humans.
Collaborative robots that work alongside humans in fields like healthcare, adapting their tasks based on the needs of the patients.

Self-Aware AI : Self-Aware AI is an advanced stage of AI that possesses self-consciousness and awareness. This type of AI would have the ability to not only understand and react to emotions but also have its own consciousness, similar to human awareness.
While we are far from achieving self-aware AI, it remains the ultimate goal for AI development. It opens philosophical debates about consciousness, identity, and the rights of AI systems if they ever reach this level.
Potential Applications:
Fully autonomous systems that can make moral and ethical decisions.
AI systems that can independently pursue goals based on their understanding of the world around them.

Role of AI

Autonomous Vehicles:

AI plays a pivotal role in autonomous vehicles by enabling them to perceive, interpret, and respond to their environment in real-time. Through sensors such as cameras, LiDAR, and radar, AI processes vast amounts of data to identify objects, pedestrians, and road markings. Machine learning algorithms then analyze this data to make decisions on steering, acceleration, and braking, ensuring safe navigation. AI also facilitates pedictive capabilities, anticipating potential hazards and optimizing routes for efficiency. Continual learning and adaptation allow autonomous vehicles to improve performance over time, promising safer transportation, reduced congestion, and enhanced mobility in the future.

Remote patient monitoring:

AI revolutionizes remote patient monitoring by enhancing data analysis, early detection, and personalized care. Through wearable devices and IoT sensors, AI collects patient data like vital signs, activity levels, and symptoms in real-time. Machine learning algorithms analyze this data to detect anomalies, predict health deterioration, and recommend interventions. AI-driven predictive analytics enable healthcare providers to intervene proactively, preventing emergencies and reducing hospital readmissions. Additionally, AI assists in personalized treatment plans by considering individual patient data and medical histories. This empowers patients to manage chronic conditions effectively from home while improving healthcare outcomes and reducing costs.

Civil Construction:

AI revolutionizes civil construction by optimizing planning, design, and execution processes. AI-driven algorithms analyze vast amounts of data to predict project timelines, costs, and potential risks accurately. Machine learning enhances design by generating efficient structural models and suggesting innovative solutions. AI-enabled drones and robots perform tasks such as site inspection, surveying, and material handling with precision and speed, improving safety and productivity. Furthermore, AI aids in real-time monitoring of construction sites, detecting defects and deviations for timely intervention. By streamlining workflows and enhancing decision-making, AI transforms civil construction, leading to cost savings, faster project delivery, and infrastructure that meets the demands of modern society.

EEE- Product defect detection:

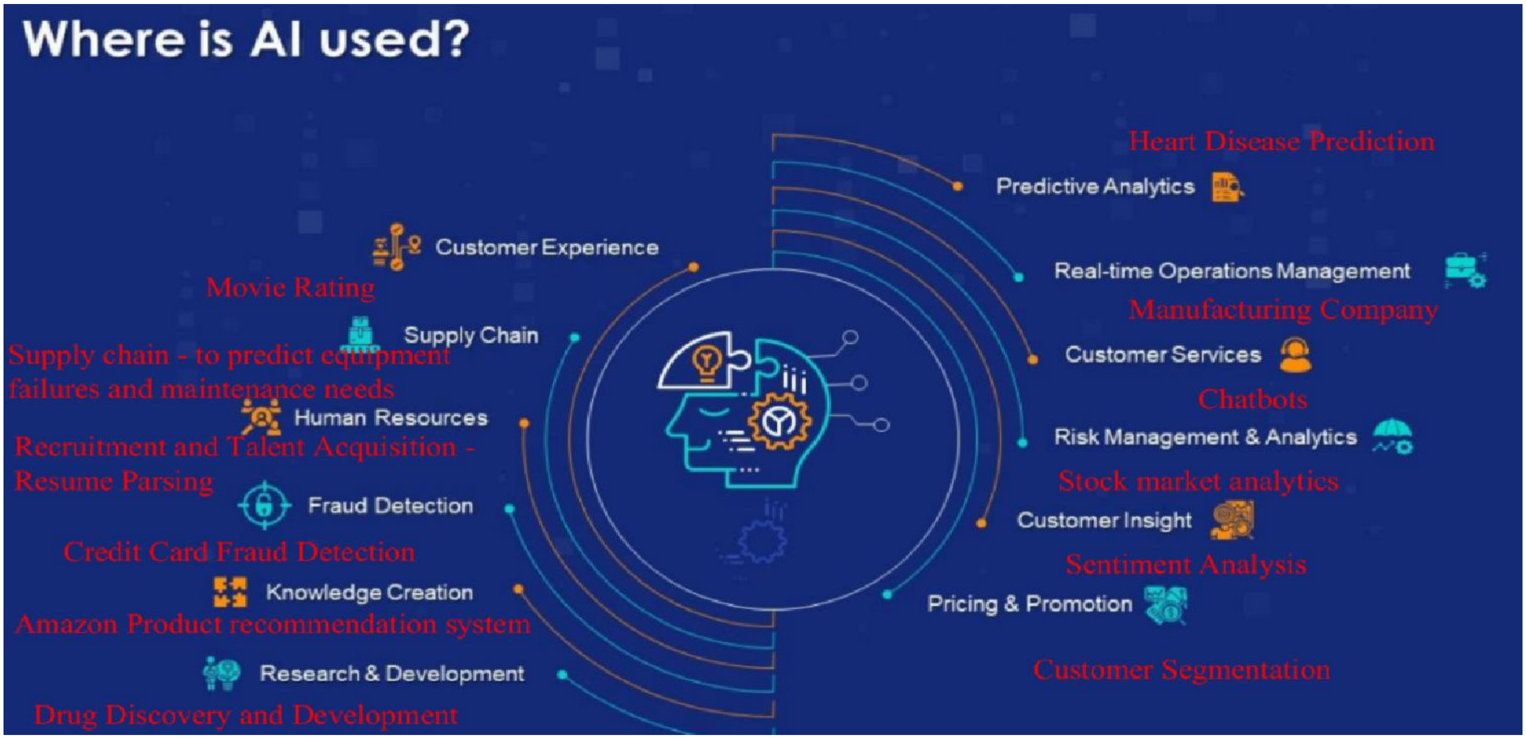
AI plays a crucial role in product defect detection in Electrical and Electronic Engineering (EEE). Through machine learning algorithms, AI analyzes vast datasets of product images or sensor data to identify defects such as faulty components, wiring errors, or manufacturing flaws. Deep learning models can classify defects with high accuracy, enabling automated quality control processes. AI-powered inspection systems detect defects in real-time during production, reducing the likelihood of defective products reaching consumers. By improving quality assurance and reducing manual inspection time, AI enhances product reliability, safety, and customer satisfaction in the EEE industry while optimizing manufacturing efficiency and reducing costs.

Smart Factory:

AI transforms smart factories by optimizing operations, predictive maintenance, and resource utilization. Through IoT sensors and connected devices, AI collects and analyzes real-time data on equipment performance, production processes, and supply chain logistics. Machine learning algorithms identify patterns, anomalies, and inefficiencies, enabling predictive maintenance to minimize downtime and prevent breakdowns. AI-driven analytics optimize production schedules, inventory management, and energy consumption, enhancing efficiency and reducing costs. Autonomous robots guided by AI perform tasks such as assembly, packaging, and material handling, improving productivity and safety. Overall, AI empowers smart factories to achieve higher levels of automation, flexibility, and responsiveness to market demands.

Telecommunication – NLP:

AI enhances telecommunications through Natural Language Processing (NLP) by automating customer interactions, improving service efficiency, and enhancing user experience. NLP-powered chatbots and virtual assistants understand and respond to customer queries in real-time, reducing wait times and improving satisfaction. AI analyzes customer feedback, sentiment, and trends from text data, enabling telecom companies to personalize services and target marketing campaigns effectively. Moreover, NLP facilitates voice recognition for voice-controlled devices and speech-to-text transcription, simplifying communication channels. By leveraging NLP, telecom providers streamline operations, gain insights into customer preferences, and deliver more responsive and tailored services, ultimately fostering stronger customer relationships and increasing competitiveness in the industry.



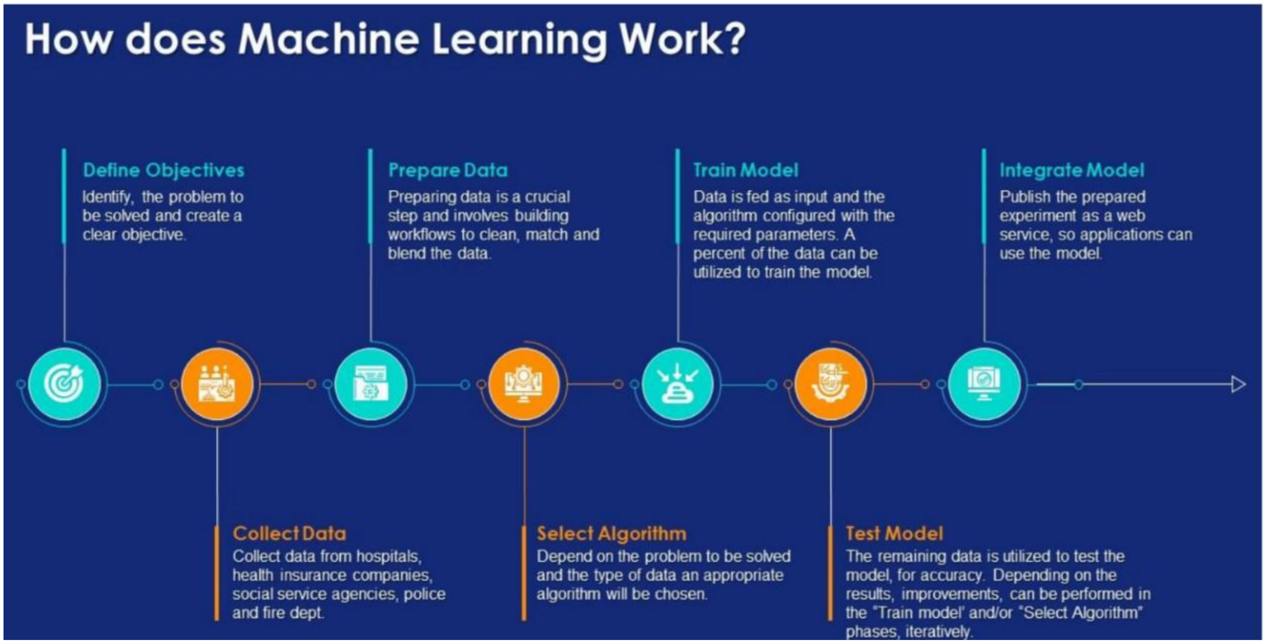
What is Machine Learning

Firstly Machine learning is a Subset of Artificial Intelligence. Machine learning is a method of data analysis that automates analytical model building. It is a branch of artificial intelligence based on the idea that systems can learn from data, identify patterns and make decisions with minimal human intervention.

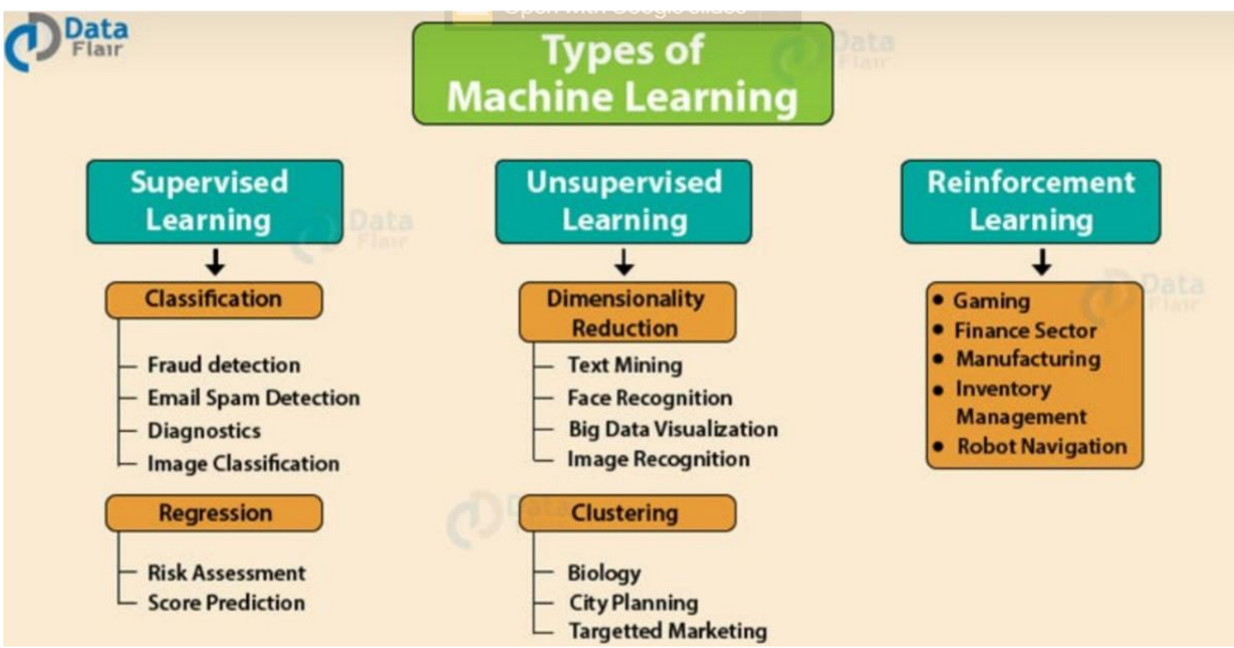
Do you get automatic recommendations on Netflix and Amazon Prime about the movies you should watch next? Or maybe you get options for People You may know on Facebook or LinkedIn? You might also use Siri, Alexa, etc. on your phones. That's all Machine Learning! This is a technology that is becoming more and more popular. Chances are that Machine Learning is used in almost every technology around you!

Machine learning is a subfield of artificial intelligence that uses algorithms trained on data sets to create models that enable machines to perform tasks that would otherwise only be possible for humans, such as categorizing images, analyzing data, or predicting price fluctuations.

In common usage, the terms "machine learning" and "artificial intelligence" are often used interchangeably with one another due to the prevalence of machine learning for AI purposes in the world today. But, the two terms are meaningfully distinct. While AI refers to the general attempt to create machines capable of human-like cognitive abilities, machine learning specifically refers to the use of algorithms and data sets to do so.



Types of Machine Learning



1. Supervised Machine Learning:

As its name suggests, Supervised machine learning is based on supervision. It means in the supervised learning technique, we train the machines using the "labeled" dataset, and based on the training, the machine predicts the output. Here, the labeled data specifies that some of the inputs are already mapped to the output. More preciously, we can say; first, we train the machine with the input and corresponding output, and then we ask the machine to predict the output using the test dataset.

Let's understand supervised learning with an example. Suppose we have an input dataset of cats and dog images. So, first, we will provide the training to the machine to understand the images, such as the shape & size of the tail of cat and dog, Shape of eyes, colour, height (dogs are taller, cats are smaller), etc. After completion of training, we input the picture of a cat and ask the machine to identify the object and predict the output. Now, the machine is well trained, so it will check all the features of the object, such as height, shape, color, eyes, ears, tail, etc., and find that it's a cat. So, it will put it in the Cat category. This is the process of how the machine identifies the objects in Supervised Learning.

Advantages:

- Since supervised learning work with the labeled dataset so we can have an exact idea about the classes of objects.
- These algorithms are helpful in predicting the output on the basis of prior experience.

Disadvantages:

- These algorithms are not able to solve complex tasks.
- It may predict the wrong output if the test data is different from the training data.
- It requires lots of computational time to train the algorithm.

2. Unsupervised Machine Learning:

Unsupervised learning is different from the Supervised learning technique; as its name suggests, there is no need for supervision. It means, in unsupervised machine learning, the machine is trained using the unlabeled dataset, and the machine predicts the output without any supervision.

In unsupervised learning, the models are trained with the data that is neither classified nor labeled, and the model acts on that data without any supervision.

The main aim of the unsupervised learning algorithm is to group or categories the unsorted dataset according to the similarities, patterns, and differences. Machines are instructed to find the hidden patterns from the input dataset.

Let's take an example to understand it more preciously; suppose there is a basket of fruit images, and we input it into the machine learning model. The images are totally unknown to the model, and the task of the machine is to find the patterns and categories of the objects.

So, now the machine will discover its patterns and differences, such as color difference, shape difference, and predict the output when it is tested with the test dataset.

Advantages:

- These algorithms can be used for complicated tasks compared to the supervised ones because these algorithms work on the unlabeled dataset.
- Unsupervised algorithms are preferable for various tasks as getting the unlabeled dataset is easier as compared to the labeled dataset.

Disadvantages:

- The output of an unsupervised algorithm can be less accurate as the dataset is not labeled, and algorithms are not trained with the exact output in prior.
- Working with Unsupervised learning is more difficult as it works with the unlabeled dataset that does not map with the output.

4. Reinforcement Learning

Reinforcement learning works on a feedback-based process, in which an AI agent (A software component) automatically explore its surrounding by hitting & trail, taking action, learning from experiences, and improving its performance. Agent gets rewarded for each good action and gets punished for each bad action; hence the goal of reinforcement learning agent is to maximize the rewards.

In reinforcement learning, there is no labeled data like supervised learning, and agents learn from their experiences only.

The reinforcement learning process is similar to a human being; for example, a child learns various things by experiences in his day-to-day life. An example of reinforcement learning is to play a game, where the Game is the environment, moves of an agent at each step define states, and the goal of the agent is to get a high score. Agent receives feedback in terms of punishment and rewards.

Due to its way of working, reinforcement learning is employed in different fields such as Game theory, Operation Research, Information theory, multi-agent systems.

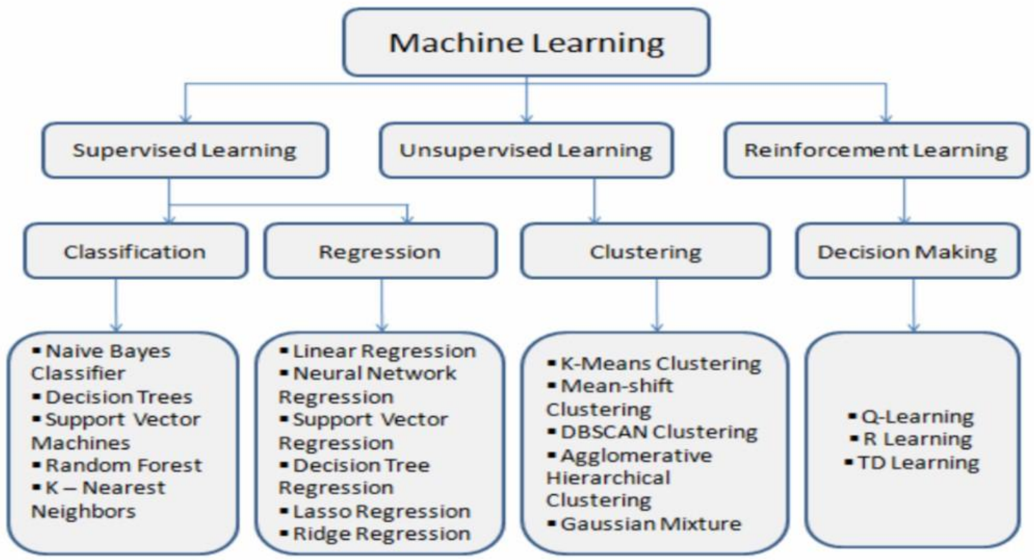
A reinforcement learning problem can be formalized using Markov Decision Process(MDP). In MDP, the agent constantly interacts with the environment and performs actions; at each action, the environment responds and generates a new state.

Advantages

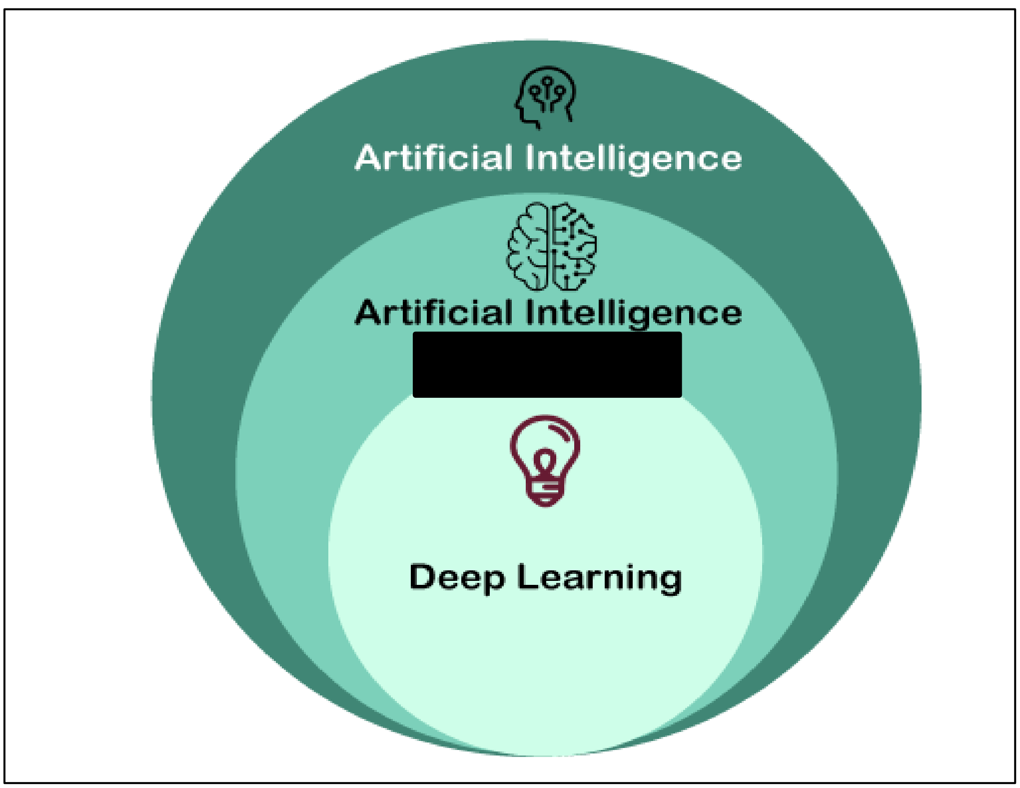
- It helps in solving complex real-world problems which are difficult to be solved by general techniques.
- The learning model of RL is similar to the learning of human beings; hence most accurate results can be found.
- Helps in achieving long term results.

Disadvantage

- RL algorithms are not preferred for simple problems.
- RL algorithms require huge data and computations.
- Too much reinforcement learning can lead to an overload of states which can weaken the results.



Difference between Machine Learning and Deep Learning



Parameter	Machine Learning	Deep Learning
Data Dependency	Although machine learning depends on the huge amount of data, it can work with a smaller amount of data.	Deep Learning algorithms highly depend on a large amount of data, so we need to feed a large amount of data for good performance.
Execution time	Machine learning algorithm takes less time to train the model than deep learning, but it takes a long-time duration to test the model.	Deep Learning takes a long execution time to train the model, but less time to test the model.
Hardware Dependencies	Since machine learning models do not need much amount of data, so they can work on low-end machines.	The deep learning model needs a huge amount of data to work efficiently, so they need GPU's and hence the high-end machine.
Feature Engineering	Machine learning models need a step of feature extraction by the expert, and then it proceeds further.	Deep learning is the enhanced version of machine learning, so it does not need to develop the feature extractor for each problem; instead, it tries to learn high-level features from the data on its own.
Problem-solving approach	To solve a given problem, the traditional ML model breaks the problem in sub-parts, and after solving each part, produces the final result.	The problem-solving approach of a deep learning model is different from the traditional ML model, as it takes input for a given problem, and produce the end result. Hence it follows the end-to-end approach.
Interpretation of result	The interpretation of the result for a given problem is easy. As when we work with machine learning, we can interpret the result easily, it means why this result occur, what was the process.	The interpretation of the result for a given problem is very difficult. As when we work with the deep learning model, we may get a better result for a given problem than the machine learning model, but we cannot find why this particular outcome occurred, and the reasoning.
Type of data	Machine learning models mostly require data in a structured form.	Deep Learning models can work with structured and unstructured data both as they rely on the layers of the Artificial neural network.
Suitable for	Machine learning models are suitable for solving simple or bit-complex problems.	Deep learning models are suitable for solving complex problems.

Unit 2: AI Project Cycle

Introduction to AI Project Cycle

The **AI Project Cycle** is a structured approach used to build AI-based solutions. It ensures that AI systems are developed systematically, considering data, models, and business goals.

Key Stages of the AI Project Cycle:

1. **Problem Scoping** – Identifying and understanding the problem.
2. **Data Acquisition** – Gathering the required data.
3. **Data Exploration** – Analyzing and preparing data for model training.
4. **Data Visualization** – Representing data using graphs and charts.
5. **Model Building & Evaluation** – Developing, testing, and optimizing AI models.

Each of these steps is crucial in developing an AI solution that is **accurate, efficient, and relevant** to real-world problems.

1. Understanding Problem Scoping

What is Problem Scoping?

Problem scoping is the **first and most crucial step** in any AI project. It involves defining the problem clearly and identifying key factors that impact the solution.

Why is it Important?

- Ensures that AI models solve the **right problem**.
- Helps in selecting **relevant data**.

- Defines project **goals, stakeholders, and constraints**.

Steps in Problem Scoping:

1. **Identify the Problem** – What issue needs to be solved?
2. **Understand Stakeholders** – Who will benefit from the solution?
3. **Analyze Constraints** – What are the limitations (data availability, resources, time, etc.)?
4. **Define the Success Metrics** – How will we measure success?

Example: Predicting Student Performance

Let's say a university wants to predict student performance based on study hours, past scores, and attendance.

- **Problem Statement:** Predict student grades based on various factors.
- **Stakeholders:** Students, teachers, university management.
- **Constraints:** Limited data, variations in grading patterns.
- **Success Metrics:** Accuracy of predictions, improvement in student results.

2. Data Acquisition

What is Data Acquisition?

Data acquisition is the process of **collecting and gathering** relevant data for AI models. This step ensures that the model has enough information to learn patterns and make predictions.

Sources of Data:

1. **Public Datasets** – Kaggle, UCI Machine Learning Repository.

2. **APIs** – Twitter API for sentiment analysis, Google Maps API for location data.
3. **Web Scraping** – Extracting data from websites using BeautifulSoup or Scrapy.
4. **User-Generated Data** – Surveys, IoT devices, sensor readings.

Example: Collecting Data for Student Performance Prediction

- Data can be collected from school records, attendance sheets, and test scores.
- A dataset may include columns like **Student ID, Study Hours, Past Grades, Attendance, and Final Score**.

3. Data Exploration

What is Data Exploration?

Data exploration is the **process of analyzing and understanding the dataset** before using it for AI modeling. This includes:

- Checking for **missing values**.
- Understanding **data types and distributions**.
- Identifying **outliers and inconsistencies**.

Why is Data Exploration Important?

- Helps in **cleaning the data** before training models.
- Identifies **patterns and relationships** between variables.
- Ensures that **incorrect or misleading data** does not affect AI predictions.

. Data Visualization

What is Data Visualization?

Data visualization helps in representing **complex datasets using graphs and charts**. It makes it easier to **identify patterns, trends, and relationships** between variables.

Types of Visualizations in AI Projects:

1. **Bar Charts** – Used for comparing categories.
2. **Histograms** – Used for understanding distributions of data.
3. **Scatter Plots** – Show relationships between two variables.
4. **Box Plots** – Help in detecting outliers in data.

Insights from Visualizations:

- If a scatter plot shows a **positive correlation** between study hours and scores, it indicates that more study hours lead to higher grades.
- A box plot can help detect **outliers** in student attendance records.

Key Takeaways:

- **A well-defined problem statement** is crucial for AI projects.
- **Data quality matters**—missing values and errors can impact AI predictions.
- **Visualization helps understand patterns** before building models.

Definition of Image Processing

Image Processing involves performing operations on images to enhance, manipulate, or analyze them. It focuses on transforming an image to extract useful information, enhance quality, or prepare it for further analysis.

Key Techniques in Image Processing:

Image enhancement (sharpening, contrast adjustment)

Noise reduction and filtering

Edge detection and segmentation

Image compression and transformation

Example:

Applying a Gaussian filter to smooth an image.

Detecting edges using Sobel or Canny operators.

Definition of Computer Vision

Computer Vision (CV) is a subfield of AI that enables machines to interpret and understand visual data from the real world. It focuses on extracting high-level information and making decisions based on images or videos.

Key Techniques in Computer Vision:

Object detection and recognition

Image classification and segmentation

Motion analysis and tracking

Facial recognition and pose estimation

Example:

Detecting pedestrians in real-time for autonomous driving.

Classifying objects in an image using Convolutional Neural Networks (CNNs).

Real-Time Scenario: Traffic Violation Detection System

Problem Statement:

The traffic police department of a metropolitan city wants to automate the process of detecting traffic violations and issuing penalties. Manual surveillance is prone to errors, and monitoring high-traffic areas in real time is challenging. They want to use CCTV footage from traffic signals to automatically identify vehicles violating traffic rules, such as:

Jumping red lights

Overspeeding

Not wearing helmets (for two-wheelers)

The goal is to:

Capture violations accurately using AI.

Recognize license plates and identify the violators.

Automate penalty issuance by integrating with the vehicle registration database.

Step 1: Problem Scoping

What is Problem Scoping?

Problem scoping is the process of defining the problem clearly and identifying key factors that impact the solution. This stage ensures that the AI model solves the correct problem.

Tasks in Problem Scoping:

Identify the Problem: Automate traffic violation detection and issue penalties.

Understand Stakeholders: Traffic police, city administration, and citizens.

Define Constraints:

High-resolution video footage required.

Real-time processing with minimal delays.

Different lighting and weather conditions may affect image quality.

Establish Success Metrics:

High accuracy in violation detection.

Low false positive rate in license plate recognition.

Faster response time for penalty issuance.

Example:

Problem Statement: Develop an AI system that detects red light violations and overspeeding from CCTV footage and automatically extracts vehicle details.

Success Criteria: The system should achieve 90%+ accuracy in detecting violations and license plate recognition.

Step 2: Data Acquisition

What is Data Acquisition?

Data acquisition involves collecting relevant data for training and testing the AI model. The quality and quantity of data directly impact the model's performance.

Sources of Data:

CCTV Footage: Real-time videos from traffic cameras at key junctions.

Public Datasets: Pre-existing datasets like:

Indian Number Plate Datasets

Open Traffic Violation Datasets from Kaggle/UCI

Annotated Data: Manual annotation of video frames to label violations.

Vehicle Registration Database: To match license plates with vehicle details.

Challenges in Data Collection:

Variability in Lighting and Weather: Low visibility during night or fog.

Motion Blur: Fast-moving vehicles may introduce blurring.

Multiple Vehicles in Frame: Need to correctly identify the violating vehicle.

Example:

Collect 50,000+ labeled images of vehicles violating traffic rules.

Annotate 10,000+ license plates manually for training an OCR model.

Step 3: Data Exploration

What is Data Exploration?

Data exploration involves analyzing and understanding the dataset to detect inconsistencies, missing values, and patterns. It prepares the data for further processing.

Tasks in Data Exploration:

Check for Missing Values: Ensure that there are no gaps in critical data.

Analyze Data Types: Confirm that numeric and categorical data types are correctly identified.

Identify Outliers: Detect and handle outliers (e.g., unrealistic speeds).

Visualize Data Patterns: Identify relationships between variables (e.g., time of day and violation frequency).

Example:

Analyze 100,000 traffic violation entries to check for incomplete or duplicate records.

Identify peak traffic hours and common violation types.

Step 4: Data Visualization

What is Data Visualization?

Data visualization involves representing data visually to detect patterns and trends that might not be obvious from raw data. This step helps identify potential insights for model development.

Common Visualization Techniques:

Bar Charts: Compare the frequency of violation types.

Histograms: Analyze the speed distribution of vehicles.

Scatter Plots: Show relationships between speed and time.

Heatmaps: Identify high-violation zones during peak hours.

Example:

Analyze violation patterns during rush hours using heatmaps.

Visualize overspeeding incidents to determine high-risk areas.

Step 5: Model Building & Evaluation

What is Model Building?

Model building involves training AI models to perform tasks such as:

Detecting vehicles and identifying violations.

Recognizing license plates using OCR.

AI Models Used:

Object Detection Model:

Model: YOLOv5

Task: Detect vehicles, traffic lights, and pedestrians.

OCR Model:

Model: Tesseract or EasyOCR

Task: Extract text from license plates.

Model Evaluation Metrics:

Precision: Percentage of correctly detected violations.

Recall: Percentage of actual violations correctly identified.

F1-Score: Balance between precision and recall.

Final Solution and Deployment

Solution Workflow:

CCTV Footage Capture: Real-time footage from traffic cameras.

Object Detection: Detect vehicles, signals, and potential violations.

License Plate Recognition: Extract vehicle details for penalty issuance.

Violation Reporting: Notify violators through automated systems.

Impact of AI Solution:

Reduced manual workload for traffic police.

Faster and more accurate identification of traffic violations.

Improved compliance with traffic laws, ensuring road safety.

Key Takeaways:

Problem Scoping: Clearly define objectives and constraints.

Data Acquisition: Collect diverse data from reliable sources.

Data Exploration and Visualization: Analyze data for patterns and trends.

Model Building: Use AI models to automate traffic violation detection.

UNIT III

Basics of Data Science

Foundations of Artificial Intelligence

Study Notes

Topics Covered:

Introduction to Data Science • Applications of Data Science • AI Project Cycle

Introduction to Data Science	Applications of Data Science	AI Project Cycle

Table of Contents

1. Introduction to Data Science

1.1 What is Data Science?

Data Science is an interdisciplinary field that uses scientific methods, algorithms, processes, and systems to extract **knowledge and insights** from structured and unstructured data. It combines elements of mathematics, statistics, computer science, and domain expertise to analyse large volumes of data and support data-driven decision-making.

■ **Key Definition**

"Data Science is the art of turning raw data into actionable insights using statistics, programming, and domain knowledge."

1.2 Key Components of Data Science

Component	Description	Tools / Skills
Statistics & Mathematics	Foundation for understanding data distributions, hypothesis testing, probability	Mean, Median, Regression, Probability
Programming	Writing code to collect, clean, and analyse data	Python, R, SQL
Machine Learning	Building models that learn from data to make predictions	scikit-learn, TensorFlow, PyTorch
Data Visualisation	Communicating findings through graphs and dashboards	Matplotlib, Tableau, Power BI
Domain Knowledge	Understanding the business/field context for meaningful analysis	Varies by industry

1.3 Data Science vs. Related Fields

It is important to distinguish Data Science from closely related fields:

Field	Focus	Relationship to Data Science
Data Science	Full pipeline: collect → analyse → predict → communicate	The umbrella field
Data Analytics	Analysing existing data to answer specific questions	Subset; focuses on past data
Machine Learning	Building self-learning algorithms	Technique used within Data Science

Artificial Intelligence	Creating intelligent behaviour in machines	Broader field; Data Science is a tool
Big Data	Storage and processing of massive datasets	Infrastructure that feeds Data Science

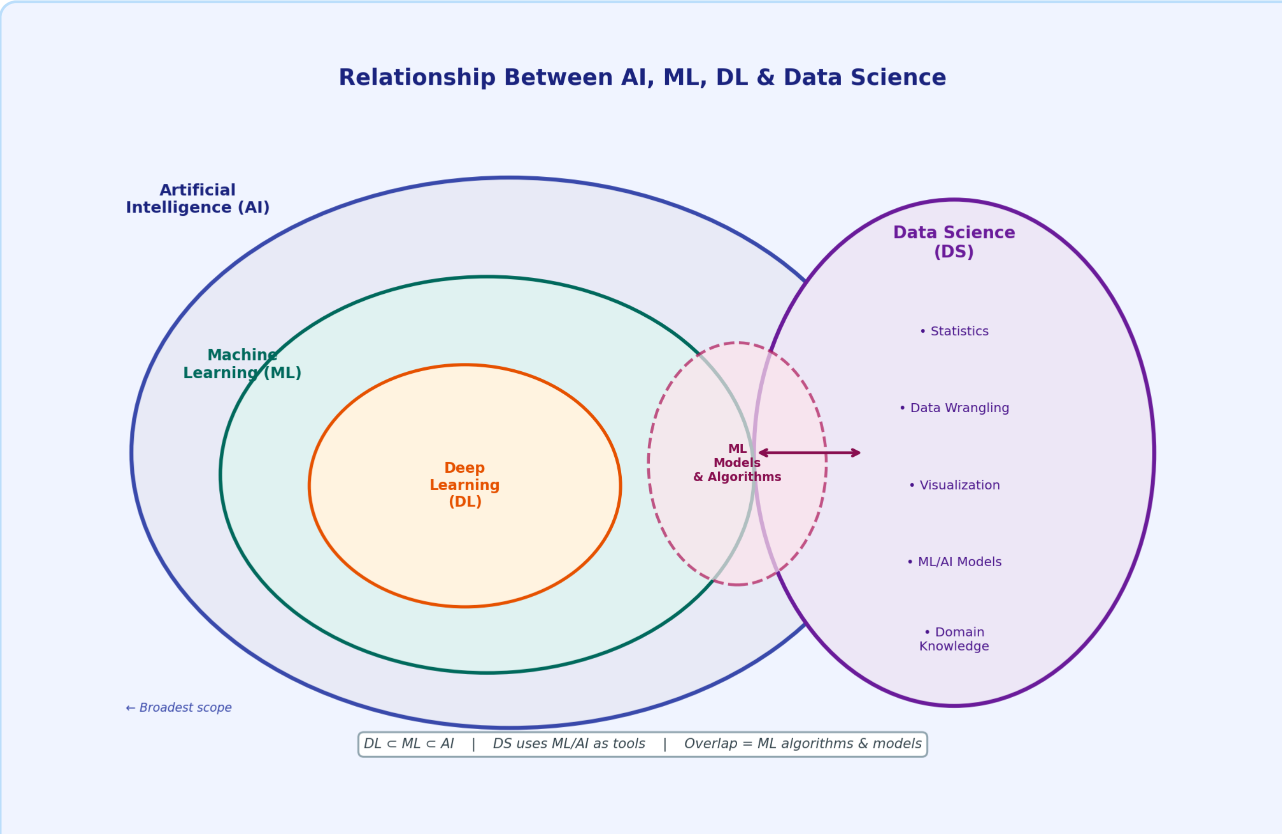


Figure 1: Venn diagram showing the relationship between AI, ML, DL and Data Science. DL is a subset of ML, which is a subset of AI. Data Science is a broader field that uses ML/AI as tools, overlapping in the area of algorithms and models.

1.4 Data Science Workflow (Step-by-Step)

A typical Data Science project follows these ordered stages:

Step 1	**Problem Definition** Clearly state the question you want to answer.	*Example: e.g. "Can we predict whether a customer will leave next month?"*
Step 2	**Data Collection** Gather relevant data from databases, APIs, web scraping, sensors, surveys.	*Example: e.g. Customer transaction history, support call logs*
Step 3	**Data Cleaning** Handle missing values, remove duplicates, fix errors (also called Data Wrangling).	*Example: e.g. Filling missing ages with median age*

Step 4	**Exploratory Data Analysis (EDA)** Summarise and visualise data to find patterns, outliers, correlations.	*Example: e.g. Histogram of customer ages, correlation heatmap*
Step 5	**Model Building** Choose and train a machine learning algorithm on the data.	*Example: e.g. Training a Random Forest classifier*
Step 6	**Model Evaluation** Test model accuracy on unseen data using metrics like accuracy, F1-score.	*Example: e.g. Achieving 87% accuracy on test set*
Step 7	**Deployment & Communication** Deploy the model and present insights to stakeholders.	*Example: e.g. Dashboard showing churn risk per customer*

■ Real-World Workflow Example – Netflix Recommendation

1. **Problem:** How do we keep users watching more content?
2. **Data:** Viewing history, ratings, search terms, watch duration
3. **Analysis:** Group users by taste profile using clustering
4. **Model:** Collaborative Filtering (what similar users liked)
5. **Result:** 80% of Netflix content watched comes from recommendations

2. Applications of Data Science

Data Science is transforming every major industry. Below are the most important domains with concrete real-world examples:

2.1 Healthcare & Medicine

Data Science helps doctors diagnose diseases faster, personalise treatment, and manage hospitals efficiently.

- **Disease Prediction:** ML models analyse patient data to predict diabetes, cancer, or heart disease risk.
- **Medical Imaging:** Deep learning detects tumours in X-rays and MRI scans with accuracy matching radiologists.
- **Drug Discovery:** Data Science accelerates identification of drug candidates from millions of compounds.
- **Hospital Management:** Predict patient admission rates to allocate beds and staff optimally.
- **Epidemic Forecasting:** Track and predict spread of diseases (e.g. COVID-19 modelling).

■ Google DeepMind – Eye Disease Detection

DeepMind's AI analysed 3D retinal scans from over 14,000 patients and identified more than 50 eye diseases with accuracy comparable to expert ophthalmologists. Early detection prevents blindness in millions.

2.2 Finance & Banking

Financial institutions leverage data science for risk management, fraud detection, and personalised banking.

- **Fraud Detection:** Real-time transaction scoring flags suspicious activities (credit card fraud).
- **Credit Scoring:** ML models replace traditional credit scoring with more accurate risk assessment.
- **Algorithmic Trading:** Automated systems execute thousands of trades per second based on data patterns.
- **Customer Churn Prediction:** Banks identify customers likely to leave and intervene proactively.
- **Loan Approval:** Automated systems assess loan applications in seconds using 100s of data points.

■ PayPal – Fraud Prevention

PayPal processes 15 million transactions per day. Their data science models analyse each transaction in under 1 millisecond across 200+ risk variables, reducing fraud losses by over 50% while maintaining smooth user experience.

2.3 E-Commerce & Retail

- **Recommendation Engines:** "Customers who bought X also bought Y" – drives 35% of Amazon's revenue.

- **Dynamic Pricing:** Amazon changes prices up to 2.5 million times per day based on demand data.
- **Inventory Management:** Predict stock requirements to avoid stockouts or overstock.
- **Customer Sentiment Analysis:** NLP on reviews to understand customer satisfaction.
- **Personalised Marketing:** Targeted ads and emails based on browsing and purchase history.

2.4 Transportation & Logistics

- **Route Optimisation:** FedEx saves millions in fuel by optimising delivery routes with data.
- **Self-Driving Cars:** Tesla processes sensor, camera, and radar data in real-time for autonomous driving.
- **Predictive Maintenance:** Airlines predict engine failure before it happens to prevent accidents.
- **Traffic Management:** Smart city systems use data to adjust traffic signals dynamically.
- **Ride-Sharing Surge Pricing:** Uber uses real-time demand/supply data to set prices.

2.5 Other Key Domains

Domain	Data Science Application
Education	Personalised learning paths, dropout prediction (e.g. Coursera, Khan Academy)
Agriculture	Crop yield prediction, soil analysis, smart irrigation systems
Sports	Player performance analytics, injury prediction, game strategy (Moneyball)
Social Media	Content recommendation, sentiment analysis, fake news detection
Energy	Smart grid optimisation, energy consumption forecasting
Government	Crime prediction, disaster management, public health monitoring

■ **Why is Data Science so powerful?**

Data Science can **turn any large dataset into a competitive advantage**. With enough quality data, organisations can make faster, more accurate, and more cost-effective decisions than ever before.

3. Revisiting the AI Project Cycle

The AI Project Cycle is a systematic framework for planning, developing, and deploying an AI solution. Data Science plays a critical role in almost every stage of this cycle. Understanding this cycle helps teams avoid common pitfalls and deliver successful AI projects.

3.1 The 6 Stages of the AI Project Cycle

Stage 1 Problem Scoping

Define the problem clearly. Identify goals, stakeholders, success metrics, and constraints.

Key Activities:

- What question are we trying to answer?
- Who will use this AI system?
- What does success look like? (e.g. 90% accuracy)
- What are the ethical and legal constraints?

Example (Problem Scoping)

A hospital wants to reduce patient readmissions by 20% within 6 months.

Stage 2 Data Acquisition

Identify, collect, and store the relevant data required for the problem.

Key Activities:

- Internal data (databases, logs, records)
- External data (APIs, public datasets, web scraping)
- Primary data collection (surveys, sensors)
- Ensure data quality, volume, and variety

Example (Data Acquisition)

Collect 5 years of patient records, treatment data, and discharge notes.

Stage 3 Data Exploration & Preparation

Understand the data deeply and prepare it for modelling.

Key Activities:

- Exploratory Data Analysis (EDA) – visualise distributions
- Handle missing values (impute or remove)
- Feature engineering – create new useful features
- Data normalisation and encoding
- Split into training, validation, and test sets

Example (Data Exploration & Preparation)

Visualise readmission rates by age group; encode diagnosis codes numerically.

Stage 4 Modelling

Choose appropriate AI/ML algorithms and train models on prepared data.

Key Activities:

- Select algorithm (Decision Tree, Neural Network, SVM, etc.)
- Train the model on training data
- Tune hyperparameters for best performance
- Compare multiple models and select the best

■ Example (Modelling)

Train a Gradient Boosting classifier on patient data.

Stage 5 Evaluation

Rigorously test the model to ensure it generalises well and meets project goals.

Key Activities:

- Test on unseen data (test set)
- Compute metrics: Accuracy, Precision, Recall, F1-Score, AUC-ROC
- Check for bias and fairness
- Validate against real-world requirements

■ Example (Evaluation)

Model achieves 88% recall in identifying high-risk patients.

Stage 6 Deployment & Monitoring

Integrate the model into production and continuously monitor its performance.

Key Activities:

- API integration into existing software
- Real-time vs. batch prediction setup
- Monitor for model drift (performance degradation over time)
- Retrain model periodically with fresh data
- Collect user feedback and iterate

■ Example (Deployment & Monitoring)

Deploy as a dashboard for nurses; retrain every 3 months with new data.

3.2 How Data Science Fits in the AI Project Cycle

Data Science is not just one part of the AI Project Cycle — it underpins the entire process:

AI Project Stage	Data Science Role

Problem Scoping	Statistical analysis to confirm problem is data-solvable
Data Acquisition	Data engineering, database management, API integration
Data Preparation	Core Data Science task: EDA, cleaning, feature engineering
Modelling	Machine Learning (a key subset of Data Science)
Evaluation	Statistical metrics, A/B testing, bias analysis
Deployment	MLOps, monitoring dashboards, data pipelines for retraining

4. Summary

Data Science	An interdisciplinary field combining statistics, programming, ML, and domain expertise to extract insights from data.
Key Components	Statistics, Programming (Python/R), Machine Learning, Data Visualisation, and Domain Knowledge.
Workflow	7 steps: Problem Definition → Data Collection → Cleaning → EDA → Modelling → Evaluation → Deployment.
Healthcare	Disease prediction, medical imaging AI, drug discovery, epidemic modelling.
Finance	Fraud detection, credit scoring, algorithmic trading, churn prediction.
E-Commerce	Recommendation engines, dynamic pricing, personalised marketing, sentiment analysis.
AI Project Cycle	6 stages: Problem Scoping → Data Acquisition → Data Preparation → Modelling → Evaluation → Deployment.
DS in AI Cycle	Data Science is the backbone of the AI Project Cycle — involved in all 6 stages.

5. Practice Questions

PART A — Short Answer Questions (2 Marks)

Q1.
Define Data Science in your own words.

Q2.
List any four components of Data Science.

Q3.
What is Exploratory Data Analysis (EDA)?

Q4.
Differentiate between Data Science and Data Analytics.

Q5.
Name two applications of Data Science in healthcare.

Q6.
What is meant by "model drift" in the deployment stage?

Q7.
State the importance of domain knowledge in Data Science.

Q8.
What is the role of feature engineering in a Data Science project?

PART B — Medium Answer Questions (5 Marks)

Q1.
Explain the Data Science workflow with a suitable real-world example for each step.

Q2.
Describe five applications of Data Science in the finance and banking sector.

Q3.
Compare and contrast Data Science, Machine Learning, and Artificial Intelligence with examples.

Q4.
Explain the stages of the AI Project Cycle and how Data Science is involved at each stage.

Q5.
What is EDA? Why is it an essential step in any Data Science project? Explain with an example.

PART C — Long Answer Questions (10 Marks)

Q1.
Discuss in detail the applications of Data Science across any four major industries. For each industry, provide at least two specific use cases with examples.

Q2.
Explain the complete AI Project Cycle in detail. How does Data Science contribute to each stage? Illustrate your answer with a complete case study of your choice.

Q3.

What is Data Science? Explain its key components in detail. Describe the Data Science workflow from problem definition to deployment with a healthcare example.

U25CY201 - ENVIRONMENTAL SCIENCE AND SUSTAINABILITY LABORATORY

SHORT PROCEDURE

EXP – 1 - DETERMINATION OF TOTAL, PERMANENT AND

TEMPORARY HARDNESS OF WATER SAMPLE(EDTA METHOD)

AIM

To determine the total, permanent and temporary hardness in the given water sample by EDTA method.

SHORT PROCEDURE

TITRATION 1 : Estimation of Total hardness

BURETTE SOLUTION – EDTA solution

PIPETTE SOLUTION – 20 ml std. hard water

BUFFER SOLUTION – 5 ml ammonia buffer

INDICATOR – 4 drops of EBT

END POINT – Wine-red colour to steel blue colour

TITRATION 2 : Estimation of Permanent hardness

BURETTE SOLUTION – EDTA solution

PIPETTE SOLUTION – 20 ml of boiled hard water sample

BUFFER SOLUTION – 5 ml ammonia buffer

INDICATOR – 4 drops of EBT

END POINT – Wine-red colour to steel blue colour

Temporary Hardness: Total hardness- Permanent hardness

RESULT

Hardness present in the given sample of water

1. Temporary hardness = …………………………… ppm.
2. Permanent hardness = ……………………..…….. ppm.
3. Total hardness = …………………………….. ppm.

EXP – 2 -ARGENTOMETRIC DETERMINATION OF AVAILABLE CHLORINE IN A SAMPLE OF BLEACHING POWDER [MOHR'S METHOD]

AIM

To estimate the amount of chloride ion present in the given water sample by argentometric method.

SHORT PROCEDURE

BURETTE SOLUTION – Std. Silver Nitrate

PIPETTE SOLUTION – 20 ml of sample water

INDICATOR – 1 ml of 2 % K2CrO4

END POINT – Colour changes from yellow to reddish brown followed by

coagulation of white precipitate

RESULT

Amount of chloride ion present in the whole of the given water sample=__________ mg/l

EXP - 3 - ESTIMATION OF DISSOLVED OXYGEN IN WATER (WINKLER'S METHOD)

AIM

To determine the amount of dissolved oxygen present in the given water sample by iodometric (Winkler's) method.

SHORT PROCEDURE

BURETTE SOLUTION – Std. Sodium thiosulphate

PIPETTE SOLUTION –. 250 ml sample water (Iodine flask) to 100 ml sample water

ADDITIONAL SOLUTION -1 mL of manganous sulphate + 1 mL of alkaline potassium iodide is added to form a brown precipitate

It is allowed to settle for 15 min and then acidified with 1 mL of concentrated sulfuric acid to dissolve the precipitate and liberate iodine, after which a known volume ,100 mL of the solution is titrated with standard sodium thiosulfate solution until pale yellow (about 10 mL), starch indicator is added to produce a blue colour

INDICATOR – 4 drops of Starch

END POINT – Colour changes from blue to dark green

RESULT

The dissolved oxygen content of the given water sample is expressed in
= ____________________mg/L (ppm).

EXP – 4 - DETERMINATION OF AMMONIUM ESTIMATION — COLORIMETRIC TESTS FOR NITRATE/AMMONIUM TO STUDY N AVAILABILITY

AIM

To estimate the ammonium content present in the given sample by colorimetric method in order to study nitrogen availability.

SHORT PROCEDURE

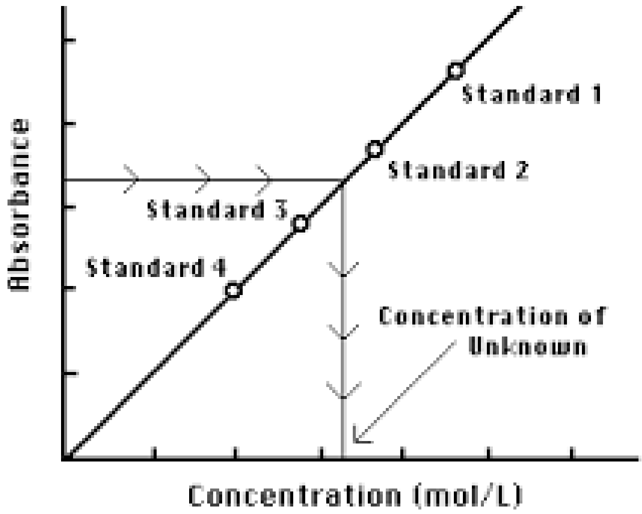
Preparation of standard curve:

1. Prepare a series of standard nitrate solutions of known concentrations (e.g., 1, 2, 3, 4, 5 ppm).
2. Preparation of samples:
    a. 5ml ammonia extract + 1 ml phenol+ 1 ml sodium nitroprusside + 2 ml of alkaline hypochlorite.
3. Set 610 nm in the calorimeter
4. Measure absorbance of each standard solutions
5. Plot a graph of absorbance (Y-axis) versus nitrate concentration (X-axis).

Determination of nitrate concentration in the sample:

1. Note the absorbance of the sample solution.
2. From the standard curve, find the corresponding unknown nitrate concentration.

S.No	Concentration	Absorbance



RESULT

The ammonium content of the given sample was found to be _____________ mg/L, indicating the availability of nitrogen in the sample.

EXP – 5- RECOVERY OF ALUMINIUM FROM WASTE MATERIALS

AIM

To recover aluminium from waste aluminium materials such as aluminium foil or scrap by chemical treatment and to demonstrate the principle of recycling and resource conservation.

SHORT PROCEDURE

1. The waste aluminium pieces are first cleaned thoroughly with water to remove grease and dirt, then dried and cut into small pieces.
2. A known mass of the aluminium waste (100mg)is taken in a beaker, and sodium hydroxide solution(5ml) is added slowly to it.
3. The mixture is heated gently until the aluminium dissolves completely, forming a clear solution with the evolution of hydrogen gas.
4. The hot solution is then filtered to remove any insoluble impurities.
5. The clear filtrate containing sodium aluminate is allowed to cool, after which dilute hydrochloric acid (5ml) is added slowly with constant stirring until a white gelatinous precipitate of aluminium hydroxide is formed.

6. The precipitate is allowed to settle, filtered, and washed thoroughly with distilled water to remove soluble impurities.
7. The aluminium hydroxide precipitate thus obtained is dried and heated in a china dish to convert it into aluminium oxide.
8. Finally, the product is cooled to room temperature and weighed.

RESULT

Aluminium was successfully recovered from the given waste material.
The percentage recovery of aluminium is **______ %**.

EXP – 6- PHOTOCATALYTIC DEGRADATION OF DYE USING TiO2

AIM

To study the photocatalytic degradation of a dye solution using titanium dioxide (TiO_2) as a photocatalyst under UV/visible light irradiation.

SHORT PROCEDURE

1. A known concentration of dye solutions (1%, 2%,3%,4%) are prepared
2. Take 1 ml of each sample in a clean beaker and 50 ml distilled water is added
3. A measured amount of TiO_2 powder(0.08g) is added to the dye solution and the mixture is stirred well for 10 minutes to ensure uniform dispersion of the catalyst.
4. The mixture is then exposed to UV llih=ght in the photo reactor
5. At regular time intervals, small aliquots of the reaction mixture are withdrawn and the TiO_2 particles are removed by filtration or centrifugation.
6. The clear solution is analyzed for absorbance at the maximum wavelength of the dye using a colorimeter
7. The decrease in absorbance with time indicates the degradation of the dye.

Calculation

The percentage degradation of the dye is calculated using the formula:

$$\% \text{ Degradation } = \frac{A_0 - A_t}{A_0} \times 100$$

Where:

A_0 = initial absorbance of the dye solution

A_t = absorbance at time t

RESULT

The photocatalytic degradation of the dye using TiO_2 was successfully studied. A gradual decrease in absorbance was observed with irradiation time, confirming the effective degradation of the dye. The percentage degradation of the dye was found to be ______ **%**.

EXP- 7 - SYNTHESIS OF BIODIESEL FROM VEGETABLE OIL.

AIM

To synthesize biodiesel from vegetable oil by transesterification process and to study an eco-friendly alternative fuel.

SHORT PROCEDURE

1. A measured volume of vegetable oil (50ml) is taken in a clean beaker or round-bottom flask and heated gently to about 55–60 °C to remove moisture.
2. A known amount of sodium hydroxide(10ml) is dissolved completely in methanol (5ml)to prepare sodium methoxide.
3. The sodium methoxide solution is slowly added to the warm vegetable oil with continuous stirring.
4. The reaction mixture is maintained at 55–60 °C and stirred for about 45–60 minutes to ensure complete transesterification.
5. After the reaction, the mixture is transferred to a separating funnel and allowed to stand undisturbed for several hours.
6. Two distinct layers are formed, with the upper layer being biodiesel and the lower layer glycerol.
7. The lower glycerol layer is separated and discarded
8. The biodiesel layer is washed gently with warm distilled water to remove residual catalyst, methanol, and impurities.

9. The washed biodiesel is then heated gently to remove traces of water and collected.

Calculation

Let,

- Volume of vegetable oil taken = () mL
- Volume (or mass) of biodiesel obtained = () mL

Percentage yield of biodiesel is calculated as: $$\%\text{Yield} = \frac{V_2}{V_1} \times 100$$

RESULT

Biodiesel was successfully synthesized from vegetable oil by the transesterification process. The percentage yield of biodiesel obtained was ______ **%**, demonstrating the feasibility of producing renewable fuel from vegetable oil.

EXP-9- DETERMINATION OF CONDUCTIVITY AND TDS TEST FOR POTABLE WATER

AIM

To determine the electrical conductivity and total dissolved solids (TDS) of the given potable water sample and to assess its suitability for drinking purposes.

SHORT PROCEDURE

1. Take five different samples in five clean beakers.
2. The samples include drinking water, tap water, HCl, NaOH, and NaCl solutions.
3. Calibrate the conductivity meter using the standard procedure.
4. Rinse the conductivity cell thoroughly with distilled water.
5. Immerse the conductivity cell in each sample one by one.
6. Record the conductivity value once the reading stabilizes.
7. Calculate the TDS value from the conductivity using the appropriate formula.

Total Dissolved Solids (TDS):

If calculated from conductivity:

$$\text{TDS (mg/L)} = \text{Conductivity } (\mu\text{S/cm}) \times K$$

Where (K) is a conversion factor (generally 0.5–0.7).

RESULT

The conductivity and TDS of the given potable water sample were determined successfully.

- Conductivity of the water sample = ______ **µS/cm**
- TDS of the water sample = ______ **mg/L**